

Characteristic Libraries and Portfolio Decisions

Mingyang Liu
Imperial Business School

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Abstract

Repeating or rescaling existing stock characteristics can change a tuned investment rule without adding information. On a common U.S. stock panel, selective replication and theme balancing alter economic attribution and stock holdings across four portfolio methods. Payoff responses depend on the method, costs, and missing-return valuation. Stock-ridge controls isolate the induced penalty change. Actual expansion from 39 to 153 characteristics has small, uncertain predictive effects under the evaluated procedures. A simulation shows that test prediction can improve without new information. Library updates therefore require separate assessment of represented inputs, estimated exposures, and cost-dependent payoffs.

Keywords: Stock characteristics; feature redundancy; portfolio choice; return prediction; regularization; model attribution.

JEL classification: G11, G17, C52, C58.

Introduction

Asset managers routinely expand the characteristic libraries used to forecast returns and construct portfolios. An update can add useful measurements, but it also changes how an estimator weights the available inputs. Repeating a characteristic adds no observation and leaves the attainable linear payoffs unchanged. Yet it can alter a regularized estimate, even when the regularization parameter is selected again. This paper asks whether such information-preserving changes affect tuned investment procedures in practice, and traces their effects through economic attribution, stock holdings, and returns after trading costs.

Our design holds the investment method, eligible securities, and decision dates fixed while changing the library representation. We repeat the columns of one economic theme and, separately, rescale themes according to their characteristic counts. The selective-copy intervention isolates a response that cannot be attributed to new observations. For stock ridge, transported penalties reconstruct the original estimation problem and provide an exact control. We also evaluate actual additions from 39 to 153 characteristics. That comparison measures the performance of the expanded procedure; it does not decompose the effect into information gains and changes in regularization.

The empirical contribution is to follow these interventions through four tuned portfolio procedures on a common stock panel. We study blocked ridge, a training-Sharpe-weighted ridge adaptation, Universal Portfolio Shrinkage (UPSA), and the single-ridge rule in the UPSA implementation. Expanded coefficients are mapped to the same 13 economic themes and named securities. A common allocation rule scales directions using historical risk and imposes a gross-exposure ceiling. Stock-level ledgers then account for entries, liquidations, return drift, financing, and transaction-cost scenarios. This design shows what a library update changes at each stage of an investment procedure.

Theme balancing changes average covariance-share allocations by 0.20–0.27 in half- ℓ_1 distance across the four methods (Table I). The corresponding distance between target stock weights is 0.19–0.28 of their average gross exposure. These are changes in fitted exposures and investment decisions, not merely a relabeling of duplicate coefficients. They do not, by themselves, establish an investment loss or identify a superior library design.

The payoff consequences depend on the method and trading assumptions. For UPSA, balancing raises the gross Sharpe ratio from 1.126 to 1.141, while the annual mean return under a 25-basis-point one-way cost falls from 1.10% to 0.92%. The paired difference is about -0.19 percentage points, with a 95% interval of $[-0.57, 0.19]$. Some selective interventions have clearer adverse effects: the seasonality-copy intervention in Sharpe-weighted ridge has a negative mean-return difference even under its declared simultaneous comparison band.

Attribution changes therefore need to be assessed alongside the trades and payoffs they induce.

Actual additions yield a different, less decisive result. Expanding monthly ridge from 39 to 153 characteristics changes normalized prediction loss by 0.0079 percentage points, with an interval of $[-0.040, 0.068]$. The monthly historical mean has lower loss than every fixed-library, copied, and family-omitted monthly linear specification. A small annual boosting search selects the mean in 25 of 30 fits. These results describe small and imprecisely estimated performance differences under the selected procedures, rather than an absence of information in the additional variables. A Gaussian simulation reinforces the distinction: independent noisy measurements reduce Bayes risk, while exact copies can improve tuned ridge without changing Bayes risk.

The library is the 153-characteristic Jensen–Kelly–Pedersen database. We evaluate 179 monthly formation decisions from January 2011 through November 2025, with returns through December 2025. Estimation uses historical windows and common formation screens within each comparison. The database is a May 2026 extraction, and the U.S. sample was used in developing the design. Missing outcomes and execution assumptions also matter: original UPSA’s annual mean is 4.79% before fees and negative under the spread proxy; its missing-return valuation scenarios range from 0.62% to 8.98%. The evidence consequently supports a diagnostic of implemented library updates, rather than a claim of investable performance on an independently confirmed sample.

The identities underlying the controls have established antecedents. Choueifaty, Froidure, and Reynier (2013) study duplication invariance in portfolio construction, and George (2010) studies redundancy in Bayesian model weighting. Our contribution is empirical: measuring the response of selected procedures and following it into stock positions and cost-dependent returns. Theme balancing is a transparent alternative weighting rule; neither its optimality nor a uniquely correct characteristic metric follows from the experiment.

The closest portfolio comparisons are Kelly et al. (2026), Ke and Pourmohammadi (2025), and Jensen et al. (2026). UPSA already aggregates characteristic-based weights into economic themes. We hold the method fixed and vary its represented inputs, rather than compare methods on a given library. Our covariance shares and fixed allocation rule also differ from utility-based importance in an investment procedure that optimizes trading costs. The regularization mechanism is related to Kozak, Nagel, and Santosh (2020). Exact copies are distinct from the richer function classes studied by Kelly, Malamud, and Zhou (2024); our experiment does not adjudicate the broader complexity arguments in Kelly and Malamud (2025) and Nagel (2025).

The paper also differs from studies of the identity of pricing information across distinct factors, such as Huang, Julliard, and Shi (2026). Our copy intervention fixes the observations and linear payoff span by construction. It does not establish that distinct anomalies contain identical pricing information. The companion working papers study metric-indexed factor identification and inference (Liu, 2026b) and characteristic geometry in portfolio choice (Liu, 2026a). Here the object is the library supplied to a tuned estimator. A supporting IPCA comparison in the Internet Appendix illustrates how component attribution can change within a fit even when its complete forecast is unchanged.

Section I defines the interventions and their controls. Section II describes the data and estimation calendar. Sections III and IV connect representation to attribution, holdings, and returns. Section V studies actual additions and the known-information simulation. Section VI assesses uncertainty and the scope of the evidence; Section VII concludes.

I. Library Design and the Estimated Investment Rule

A. Observations, representations, and fitted rules

At formation date t , let Z_t be an $N_t \times p$ matrix of characteristic observations and y_{t+1} the vector of next-month excess returns. The formation universe of N_t securities is chosen without using future-return availability. A library specifies columns, units, and labels. Its generated information set is the collection of measurable observations. Repeating a column changes the library but leaves this information set unchanged.

Write an encoded library as $\tilde{Z}_t = Z_t A$, where $A \in \mathbb{R}^{p \times q}$ is common across a fit's dates. If A has full row rank, the vectors $Z_t b$ attainable with canonical coefficients $b \in \mathbb{R}^p$ are exactly those attainable with encoded coefficients $\theta \in \mathbb{R}^q$, because $b = A\theta$. An invertible square recoding and redundant exact copies both satisfy this condition. Adding a new characteristic can instead enlarge the linear function class. Adding a nonlinear function of existing observations can enlarge that class without adding information in the probabilistic sense.

These distinctions explain why predictive improvement alone does not identify an information gain. An expanded function class can reduce the best attainable in-sample loss but worsen a finite-sample fitted procedure. A same-span recoding can improve the procedure by changing its penalty. Our controlled interventions identify the latter possibility; the actual-addition experiment evaluates the complete response to new inputs.

B. The penalty induced by repeated columns

Let $\mathcal{T}$ denote training months and $\mathcal{O}_s$ the securities with observed targets in month s , with $n_s = |\mathcal{O}_s|$. Stock ridge solves

$$(\hat{a}, \hat{b}) = \arg \min_{a,b} \left\{ \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{\|y_{s+1, \mathcal{O}_s} - a\mathbf{1} - Z_{s, \mathcal{O}_s} b\|^2}{n_s} + \lambda b' P b \right\}, \quad (1)$$

where a is an unpenalized intercept, λ is a nonnegative scalar, and the positive penalty matrix P fixes the relative cost of slope directions. All transformations below apply to the slopes after profiling the intercept. Marginal ranking or standardization does not remove the choice of P .

Under $b = A\theta$, transporting the penalty means using $\theta' A' P A \theta$. The loss and penalty then depend on the same canonical vector, so the estimation problem is unchanged. In contrast, an identity penalty on encoded coefficients induces a different canonical penalty. For invertible A , it is $b' A^{-T} A^{-1} b$; for a full-row-rank redundant A , minimizing over all representatives gives $b'(AA')^{-1}b$.

The selective-copy case makes the mechanism explicit. Replace a characteristic by m identical columns. Their coefficients enter the prediction through $b = \sum_{j=1}^m \theta_j$, and

$$\min_{\sum_j \theta_j = b} \sum_{j=1}^m \theta_j^2 = \frac{b^2}{m}. \quad (2)$$

Equal coefficients attain the minimum. Copying one theme therefore weakens its effective penalty relative to other themes. Under an independent Gaussian coefficient prior, the same operation increases the prior variance of the aggregate coefficient. Neither interpretation establishes that the change is economically undesirable.

Uniform replication supplies a control. Copying every penalized column m times and replacing every candidate λ by $m\lambda$ restores the original candidate problems. With unchanged validation and tie selection, it restores the selected rule. Selective replication generally cannot be undone by one scalar: reversing its effect in the copied directions would also change the uncopied directions. Our comparisons allow scalar retuning, so they measure the response of the selected procedure rather than a single fixed-penalty fit.

Independent expanded stock-ridge solves verify the induced native penalty; uniform-copy and synthetic ridge checks verify the corresponding unchanged-rule controls. Extending a control to an adaptive portfolio procedure requires preserving its entire selection, combination, and normalization rule. We do not claim a separately implemented transported-UPSA procedure. The four managed-return comparisons below measure responses of their native

algorithms to the represented inputs. The Internet Appendix gives the transport proof; uniqueness is required for the canonical fit, not for redundant coefficients.

C. *Exposure and theme weighting*

The ex-ante Gram matrix $Q_t = Z_t'Z_t/N_t$ measures characteristic exposure. A coefficient change d generates mean-square stock-index change

$$\frac{\|Z_t d\|^2}{N_t} = d'Q_t d. \quad (3)$$

Euclidean coefficient distance instead uses $d'd$. For example, replacing one valuation measure by a closely related measure can produce a large coefficient change and little change in stock exposures. Equation (3) measures the resulting stock index directly.

A Gram-weighted penalty is not automatically an optimal investment rule. The characteristic Gram matrix describes cross-sectional exposures, whereas managed-return covariance describes time-series payoff risk. These objects need not order directions in the same way. We retain their separate roles in estimation, attribution, and portfolio scaling.

Theme balancing is another declared weighting choice. With k_g characteristics in theme g , scaling its columns by $\sqrt{p/(13k_g)}$ makes an identity coefficient penalty equivalent to canonical precision $13k_g/p$. Under fixed marginal units, this equalizes aggregate coefficient-prior variance across themes. It depends on the taxonomy and does not equalize information or expected profitability. Its purpose here is to measure sensitivity to a transparent alternative allocation of regularization.

D. *Attribution in fixed economic coordinates*

Partition the canonical managed payoffs into 13 themes. For a fitted coefficient vector b , let u_g denote theme g 's payoff contribution and $u = \sum_g u_g$ the total. On the historical training window, define

$$C_g = \text{Cov}(u_g, u), \quad s_g = \frac{C_g}{\text{Var}(u)}. \quad (4)$$

The contributions sum to total variance and the shares sum to one when that variance is positive. A negative share records a hedging contribution. A zero direction has undefined shares and holds cash. These allocations describe fitted payoff risk, not a theme's unique information or equilibrium risk price.

The covariance contribution equals the Shapley allocation of the specific game $v(S) = \text{Var}(\sum_{g \in S} u_g)$: each pairwise covariance is split equally between its two themes. It is therefore not an independent robustness measure. The Internet Appendix proves the identity and examines a distinct game based on projecting the total payoff onto coalitions of scalar theme

payoffs. Different games answer different attribution questions (Shapley, 1952; Sundararajan and Najmi, 2020).

We measure attribution change by $D_s = \frac{1}{2} \sum_g |s_g^{\text{new}} - s_g^{\text{original}}|$, keeping the same 13 themes after canonical mapping. Since the shares can be negative, this distance can exceed one. We compare it with changes in stock targets and cost-dependent returns. Each statistic measures a different consequence of the library intervention.

E. The empirical comparisons

The predictive estimand is the paired mean change in stock-level squared error on common observed targets and dates. The investment estimands are changes in mean excess return, volatility, Sharpe ratio, and a quadratic certainty-equivalent approximation under a stated allocation rule. Numerical transport errors instead assess whether implementations solve the same problem.

A separate Gaussian simulation gives a case in which information is known. If $h, e_1, \dots, e_m$ are independent standard normals and $x_j = h + \tau e_j$, observing m independent measurements gives $\text{Var}(h \mid x_1, \dots, x_m) = \tau^2 / (\tau^2 + m)$. Exact copies of x_1 leave the conditional distribution unchanged. So do independent noisy transformations of x_1 when x_1 remains observed. Section V uses these cases to distinguish a lower Bayes risk from a better fitted ridge rule. This known measurement model is not imposed on the actual characteristic library.

II. Data, Timing, and Library Interventions

A. Sources and formation universe

We use the U.S. annual files of the Jensen–Kelly–Pedersen database (Jensen, Kelly, and Pedersen, 2023), obtained in May 2026 through WRDS `contrib.global_factor`. The source provides 153 characteristics and their inherited 13-theme cluster labels. The labels describe the source clustering, so individual names need not match a purely semantic classification. Returns are the source’s processed WRDS fields; we do not independently rebuild return histories from raw CRSP. The replication materials identify the archived files and reconstruct the panel from them.

The estimation panel covers formation months January 2000–November 2025, with forward returns through December 2025. Eligible securities have the source’s common-stock, main-exchange, primary-security, and main-observation flags equal to one, positive finite market equity, and a size category of small, large, or mega. The smaller source categories are excluded. We retain stable security identifiers and CRSP permnos for the liquidity joins.

Eligibility, characteristic ranks, and imputation are determined before applying the next-return observation mask. Forecast loss uses the same observed targets for every model in a paired comparison. Portfolios retain stocks with missing outcomes and apply the valuation conventions in Section IV. This ordering prevents future outcome availability from determining formation holdings or other stocks' ranks.

A finite characteristic receives its average rank among finite eligible values, divided by their count plus one, minus one half. Missing values receive zero, the neutral location of this marginal scale. A constant column supplies no cross-sectional direction. The baseline treats adjacent sorted values within 10^{-13} as tied. The exact-tie sensitivity changes 122,557 of 107,195,472 encodings (0.114%), with a maximum absolute change of 0.098, and refits the affected procedures. Neutral imputation and the tie convention are part of the evaluated design; raw-value coverage and the sensitivity results are reported in the Internet Appendix and Section VI.

B. Moments and managed payoffs

Write $u_{it} = (1, z'_{it})'$ for the augmented characteristic vector, and $\beta = (a, b')'$ for its coefficient. For observed targets $i \in \mathcal{O}_t$, define

$$G_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} u_{it} u'_{it}, \quad x_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} u_{it} y_{i,t+1}, \quad q_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} y_{i,t+1}^2. \quad (5)$$

The stock-loss objective is $q_t - 2\beta'x_t + \beta'G_t\beta$, averaged equally across months. A month with more observed stocks receives no greater total weight. We separately retain the ex-ante Gram matrix $\mathcal{Q}_t = N_t^{-1} \sum_i u_{it} u'_{it}$ and outcome coverage $\pi_t = n_t/N_t$. The observed-target moment G_t determines recorded prediction loss; the ex-ante moment describes formation exposures. Direct stock calculations and their moment representation implement the same objective.

Let $\bar{z}_t$ be the mean characteristic vector among all eligible stocks. Centered characteristic weights form the managed-return vector

$$R_t = \frac{1}{N_t} \sum_i (z_{it} - \bar{z}_t) \tilde{y}_{i,t+1} = \pi_t (x_{t,z} - \bar{z}_t x_{t,0}), \quad (6)$$

where $\tilde{y}$ replaces missing vendor excess returns by zero. The subscripts z and 0 select slope and intercept entries. The equality follows by summing over the observed outcomes; it is also verified from direct stock weights. Unlike loss on observed targets, a realized portfolio payoff requires a convention for every retained position.

C. Decision timing and selection

The first evaluated formation decision is January 2011. Its 120-month training history runs from December 2000 to November 2010, with outcomes ending in December 2010. More generally, the last training formation month is $t - 2$, whose outcome is realized at $t - 1$. This common embargo excludes the immediately preceding formation month. There are 179 evaluation decisions through November 2025.

Monthly stock ridge uses 95 inner fitting months, a one-month gap, and 24 validation months. It selects among eight penalties and an intercept-only historical mean, then refits the selected specification on all 120 outer months. Exact-copy comparisons retain canonical baseline penalty units. The Internet Appendix gives the grids, intercept profiling, and endpoint-selection frequencies.

Managed blocked ridge uses the same temporal split with a different objective: validation maximizes average unscaled portfolio return minus one half its square over an absolute ridge grid from 10^{-7} to 10^3 . The Sharpe-weighted rule recalculates its precision on each training split. UPSA and its single-ridge comparator use historical leave-one-out selection within the available outer window. Every such window precedes the evaluated investment month; the inner selection protocols differ across methods.

IPCA and histogram gradient boosting refit model parameters each January and hold them fixed during that calendar year. IPCA forecast premia are historical factor averages; factors extracted from the return being forecast are used only for contemporaneous fit statistics. Its stock-risk overlay updates monthly. Annual forecasting comparisons include an annual historical mean so that a change in refit frequency is not attributed to a change in model class.

D. The interventions

The small baseline contains three characteristics per source theme, selected by a fixed name-based rule, for 39 in total. The full library contains 153. A separate historical 36-name proxy maps an established list to current vendor names; it does not recreate the earlier data vintage or sample.

The stock-ridge copy ladder starts from the 39-characteristic baseline. Repeating each of a selected theme’s three columns to two, five, or ten total copies yields encoded widths of 42, 51, and 66, respectively. We report all 39 theme–multiplicity combinations. Induced canonical penalties and independently solved expanded problems give equivalent fits. The chosen multiplicities are controlled perturbations, not estimates of duplication rates in actual signal discovery.

Actual-addition comparisons include the full library and each of its 13 theme omissions. Omissions are compared with the full library. Twelve fixed random seeds also generate stratified nested paths with 13, 39, and 78 characteristics. We report all seeds and sizes rather than select the best path. Together with the fixed libraries and copy ladder, these designs produce 93 stock-prediction specifications.

The managed-return experiment starts from all 153 payoffs and applies four methods to 15 representations: original, theme balanced, and fivefold copies of each theme separately. These 60 specifications are mapped back to canonical characteristic weights before attribution or stock allocation. Repeated coefficients are thus combined into their actual economic exposure. The experiments on real additions and exact copies remain distinct: omission or addition can change information, the function class, and the refitted procedure together.

E. Overlap, liquidity, and fixed assets

For each historical training window, we form the within-theme correlation matrix from centered ex-ante characteristic moments. If its eigenvalues are $\ell_1, \dots, \ell_k$, its participation rank is $(\sum_j \ell_j)^2 / \sum_j \ell_j^2$. This equals one for a rank-one block and k for an identity block. We report window averages of the rank and of column count divided by rank. These describe linear overlap under the chosen encoding. Strongly correlated measurements can improve precision, so the statistic does not count independent sources of predictive information.

Liquidity inputs are monthly average-dollar-volume (ADV) files constructed from WRDS `crsp.dsfs_v2` daily data and joined by permno and formation month. Coverage requires positive ADV60 with at least 60 observations. The available files do not cover every formation holding. We report trade-weighted coverage and calculate capacity only among covered trades; exits may use the last available liquidity observation.

The supporting pricing diagnostics use 75 fixed value-weighted French portfolios: 25 each sorted on size and book-to-market, operating profitability, or investment. Realization-month excess returns are aligned with six French benchmark factors. The test assets remain outside selection and unchanged across library interventions, although their underlying U.S. securities overlap with the estimation universe. The resulting intercept regressions evaluate a fixed benchmark span, not a structural price of risk.

F. Scope of the chronological evaluation

Each decision uses the stated historical window, but the design was developed on this U.S. sample and was not preregistered. The retrospective extraction does not establish point-in-time observability of every vendor field. Pairing removes sample-construction differences within an intervention; it does not cancel data errors, historical revisions, or missing-return

bias. These limits affect empirical magnitudes even though the algebraic transport identity holds on common numerical inputs.

III. Representation, Attribution, and Portfolio Performance

A. Overlap in the observed library

Table II and Figure 1 show uneven overlap across themes. Quality has 17 columns and mean correlation participation rank 3.47; low risk has 18 columns and rank 4.01; seasonality has 12 columns and rank 11.78. Their average count-to-rank ratios are 4.90, 4.49, and 1.02. Marginal ranking has not made the blocks equally dispersed in their joint characteristic space.

This pattern motivates a controlled sensitivity comparison. It does not show that quality’s real measurements are equivalent to exact copies, or that seasonality contains more useful predictive information. The next experiment changes representation while preserving the available observations and linear payoff span. The resulting response belongs to the fitted procedure.

B. Four portfolio procedures

Let $R \in \mathbb{R}^{T \times 153}$ contain the managed payoffs in a training window, with rows indexed by return month. The columns are not demeaned over time. Define $M = R'R/T$ and $\mu = R'\mathbf{1}/T$. The constant-target portfolio regression solves

$$\hat{b}(z) = \arg \min_b \left\{ T^{-1} \|\mathbf{1} - Rb\|^2 + zb'b \right\} = (M + zI)^{-1} \mu, \quad (7)$$

for positive ridge penalty z . This formulation is related to Britten-Jones (1999). It uses uncentered second moments; centered covariance is used separately for risk scaling and attribution. Stock-return prediction in equation (1) and portfolio regression in equation (7) have different dependent variables and objectives. Weak gains in the former need not imply weak gross portfolio payoffs in the latter.

Blocked ridge selects z using the temporal portfolio criterion in Section II. The Sharpe-weighted adaptation replaces I with a diagonal precision proportional to inverse squared training Sharpe ratios. Squared monthly Sharpe ratios are floored at 10^{-6} , and the diagonal vector is normalized to Euclidean norm one. Both steps are recalculated on the relevant training split. These conventions define our adaptation of Ke and Pourmohammadi (2025).

UPSA combines ridge directions using a nonnegative quadratic program fitted to leave-one-out returns (Kelly et al., 2026). We use the authors’ public implementation (Pourmohammadi, 2026), retain its trace normalization, and disable the optional sum-to-one constraint.

Portfolio weights themselves may be signed. The implementation’s selected single-ridge rule is the fourth procedure. Algorithm details and the evaluated source revision are documented in the Internet Appendix and replication materials.

Every encoded estimate is mapped back to 153 canonical coefficients before theme attribution or stock allocation. The economic taxonomy therefore always has 13 players. This differs from allocating payoffs to replicated players in a cooperative game (Han et al., 2023). The comparison measures changes in estimated economic exposure rather than treating duplicate columns as new economic entities.

C. Changes in economic composition

Table III reports the original, balanced, and fivefold value representations. Value is an illustrative familiar investment theme, not a prespecified or representative summary of all interventions. The Internet Appendix reports all 13 selective-copy cases. Mean covariance-share distance is 0.20–0.27 for balancing and 0.22–0.30 for the value example across the four methods.

For UPSA, balancing changes the mean share distance by 0.274 and value replication by 0.264. The response is consistent with applying the shrinkage algorithm to a representation that changes its sample spectrum. It does not contradict the algorithm’s portfolio-performance objective. The estimate establishes sensitivity in this application, rather than a universal ranking of shrinkage methods.

Before the stock gross ceiling, each direction with positive training volatility is scaled to a 10% annualized target. Attribution shares are unchanged by positive rescaling. The target standardizes preliminary training risk; it does not establish precise attribution or common realized risk after the gross ceiling. Share distance alone therefore has no welfare interpretation.

The projection-game sensitivity in the Internet Appendix gives nonnegative allocations in the selected cases, while covariance shares can be negative. A theme may hedge the total payoff and still help explain its variation. The two allocations differ because they ask different questions. The comparison uses the first, middle, and last evaluation dates for each original and balanced method.

D. From attribution to realized performance

The stock performance in Table IV does not follow mechanically from attribution distance. UPSA’s gross Sharpe ratios are 1.126 in the original representation, 1.141 after balancing, and 1.150 after value replication. Blocked ridge instead gives 0.366, 0.348, and 0.254. There is no uniform gain from the balanced representation.

At 25 basis points per traded dollar, balancing changes UPSA’s annual mean excess return from 1.10% to 0.92%. The paired difference of -0.19 percentage points has a pointwise 12-month-block interval of approximately $[-0.57, 0.19]$. Value replication has a positive point estimate of 0.37 percentage points, with a pointwise lower endpoint close to zero. These selected contrasts are accompanied by all-family bands and longer-block sensitivities in the Internet Appendix.

The full set includes clearer losses. For Sharpe-weighted ridge, fivefold seasonality replication changes the annual mean after 25-basis-point costs by -1.30 percentage points, with a simultaneous band of $[-2.35, -0.25]$ across its 13 selective interventions under 12-month blocks. Thus uncertainty about the headline balancing gain does not imply that every library intervention is economically indistinguishable from the original rule.

Figure 2 shows all 14 interventions per method. Its vertical axis is a training covariance-share distance; its horizontal axis is the change in realized annual mean return after the 25-basis-point cost scenario. The latter includes the stock gross ceiling, realized positions, and each strategy’s cost ledger. The scatter relates distinct consequences of the same intervention; it is not an estimated causal effect of attribution distance on returns. Section IV explains the holdings and valuation behind those payoffs.

IV. From Estimated Directions to Stock Decisions

A. A common allocation rule

A canonical slope direction b_t generates preliminary stock weights $(z_{it} - \bar{z}_t)'b_t/N_t$. Cross-sectional centering makes the active book dollar neutral. We estimate annualized volatility from the direction’s available 120-month return history, scale to a 10% target, and cap formation gross exposure at two. A direction with zero or numerically degenerate training volatility holds cash, including a mean-only forecast. The same map applies to stock-prediction and managed-return directions.

This fixed allocation makes the intervention observable in individual securities. It holds the allocation map and gross constraint constant while the estimated direction changes. It does not impose a conditional stock-covariance optimizer, industry neutrality, a 1% position cap, trade-participation limits, or short availability. A cost-aware optimizer, such as that of Jensen et al. (2026), could change the response by adjusting positions to their execution costs.

The gross ceiling also changes the risk comparison. Original UPSA has average gross exposure of two and realized annualized volatility of about 4.26%; original blocked ridge has gross exposure of 1.92 and volatility of 7.58%. Thus the common preliminary 10% target does

not yield equal realized risk. We report Sharpe ratios, realized volatility, and a quadratic certainty-equivalent approximation at a fixed risk-aversion coefficient.

B. Financing and complete trades

Previous stock holdings drift with their realized total returns and the scenario’s end-of-month net asset value (NAV). Trades compare the new target with the drifted positions over the union of old and new stock identifiers, including liquidations. The first month starts in cash. Turnover is the full sum of absolute traded dollars per pre-trade NAV. It differs from the same-date target-weight distance between two representations, which measures a change in desired exposures rather than a realized trade sequence.

Each cost scenario has its own cash account and NAV. Targets are stated per pre-trade NAV, with fees paid at the beginning of the return month. Let κ_t be the fee fraction, r_{t+1}^f the financing return, and y_{t+1} the vector of vendor excess stock returns. For a dollar-neutral target book the NAV ratio is

$$1 + r_{t+1}^f + w_t' y_{t+1} - \kappa_t(1 + r_{t+1}^f). \quad (8)$$

The last term includes the financing opportunity cost of fees. Because fees change subsequent drifted weights, each ledger is calculated recursively.

Vendor excess stock returns and French financing rates use slightly different risk-free conventions. We recover the vendor rate from same-month total minus excess returns, check that it is common across available securities, and match forward total returns to next-month observations by stock identifier. Positions drift with vendor-consistent total returns; cash financing and French test-asset comparisons use French rates. The common rate difference cancels from a new dollar-neutral book’s aggregate payoff, but affects individual position drift.

C. Trading costs and unobserved returns

We charge flat one-way costs of 10, 25, 50, or 100 basis points per traded dollar, and separately use a half-spread proxy clipped between 5 and 500 basis points. Missing current spreads use the finite cross-sectional median and then an explicit fallback; liquidation-only names use their last observed proxy. These are execution scenarios without borrow fees, calibrated market impact, or an optimized trading schedule. Figure 3 varies costs for the fixed allocation rule.

Original UPSA trades about 1.23 dollars per dollar of NAV each month. Its annual mean excess return is 4.79% before fees, 1.10% at 25 basis points, and -1.43% under the spread proxy. The original, balanced, and value-copy UPSA books all have negative means under

that proxy. Their gross Sharpe ratios therefore do not establish a profitable implementable strategy under the evaluated costs.

Positions with unobserved future returns remain in the book. The primary valuation assigns zero vendor excess return, so their total return equals the vendor risk-free rate. Separate no-fee scenarios assign -100% or $+100\%$ total returns. For original UPSA, annual mean excess returns then range from 0.62% to 8.98% , compared with 4.79% under the primary valuation. These scenarios expose sensitivity to unobserved outcomes; they are neither estimated returns nor mathematical bounds. Gross missing-outcome exposure averages about 1.3% of NAV, but its signed composition determines the payoff effect.

Average trade-weighted ADV coverage is about 97.6% for original UPSA, requiring positive ADV60 and at least 60 observations. Covered-trade capacity is the minimum of 10% of ADV divided by the absolute trade weight. Uncovered trades and exits using stale liquidity information prevent this statistic from certifying feasible assets under management for the whole book.

D. A fixed-asset pricing diagnostic

The 75 French portfolios provide a supplementary comparison on fixed assets. We estimate time-series intercepts using either a constructed stock portfolio alone or the six French factors plus that portfolio. Adding original or balanced UPSA leaves root-mean-square monthly alphas near the six-factor benchmark. The Internet Appendix reports the regression specification, all asset-level results, and the corresponding magnitudes. This exercise measures contribution to a specified benchmark span; it does not identify equilibrium prices of risk for the themes.

V. Actual Additions and a Precision Falsification

A. Prediction after adding characteristics

Actual additions change the inputs and the fitted procedure together. Unlike the exact-copy control, this comparison does not identify separate information and regularization effects. Table VI reports zero-forecast R^2 and paired loss relative to a historical mean with the same refit frequency. The monthly mean attains $R^2 = 0.239\%$, compared with 0.214% for 39 characteristics and 0.206% for 153. It has lower average loss than every fixed-library, replicated, and family-omitted monthly linear specification.

The full-minus-baseline loss difference is 0.0079 percentage points of common zero-forecast loss, with a paired 95% interval of $[-0.040, 0.068]$. The interval allows gains and losses from the expansion. Theme balancing and the Gram penalty also provide no uniform

forecasting gain. These estimates measure what the selected procedures extract; they do not identify the additional variables’ conditional information.

The stock-loss and portfolio results concern different objectives. Stock ridge minimizes individual-stock squared forecast errors. Managed-return methods combine characteristic portfolios using a constant-target regression and portfolio-based selection. A procedure that improves the latter objective need not improve the former, so the weak stock-prediction results do not negate the managed strategies’ gross returns or establish a ranking between the two approaches.

Figure 4 compares each family omission with the full library. Omitting quality raises normalized loss by about 0.026 percentage points; omitting accruals lowers it by about 0.007. Every simultaneous interval across the 13 omissions includes zero, including accruals, whose pointwise interval excludes zero. Since the remaining model is refitted and retuned, these contrasts measure the procedure’s response to removing both shared and distinct content.

Random nested libraries assess dependence on the fixed 39-name baseline. We report all 12 seeds at each of three sizes. Two of the 36 seed–size points beat the monthly mean descriptively, both at 13 characteristics. This distribution describes the specified stratified construction, with source inputs and theme labels fixed. The exact-copy ladder likewise contains improvements and deteriorations after retuning, but none survives the simultaneous band across the full set of baseline comparisons. Its role is to identify the penalty mechanism and evaluate the transported control.

B. A nonlinear benchmark with matched timing

Histogram gradient boosting compares the 39- and 153-characteristic libraries using annual January refits. Validation selects two leaf limits crossed with two iteration counts, together with a mean-only candidate. Learning rate, minimum leaf size, regularization, and random state are fixed. This limited nonlinear benchmark draws on the predictive literature (Gu, Kelly, and Xiu, 2020, 2021; Freyberger, Neuhierl, and Weber, 2020) while maintaining common data and timing.

The annual historical mean attains $R^2 = 0.319\%$. Boosting attains 0.304% for 39 characteristics and 0.326% for 153. Relative to the annual mean, the full-library normalized loss difference is -0.0063 percentage points, with interval $[-0.0623, 0.0402]$. Relative to the smaller boosting library, it is -0.0221 , with interval $[-0.0822, 0.0299]$. Neither comparison establishes a precise forecasting gain.

The mean is selected in 25 of 30 annual fits. Its coefficient, like the tree coefficients, stays fixed between January refits. Comparing boosting with the monthly mean would therefore mix model choice and update frequency. The annual-mean comparison controls for

that timing difference; its exploratory status is documented in the Internet Appendix. The results concern this grid, universe, embargo, and lookback. They do not test the full range of function-class expansions studied by Kelly, Malamud, and Zhou (2024); Didisheim et al. (2024); Kelly et al. (2025).

C. A simulation with known information

Table VII separates information gains from estimator improvements in a synthetic Gaussian model. A latent signal is measured with Gaussian noise; the outcome adds independent noise of variance 0.25. The designs supply exact copies, independent noisy measurements of the latent signal, or noisy transformations while retaining the first measurement. Each replication has independent training, validation, and test samples, common test outcomes across paired models, and eight nuisance predictors.

At measurement-noise standard deviation one, a single measurement gives Bayes outcome MSE of 0.7500. Sixteen independent measurements reduce it to 0.3088. Exact copies and conditional noisy copies leave Bayes MSE at 0.7500. Yet retuned identity ridge improves from 0.7999 to 0.7695 under 16 exact copies, while its transported counterpart preserves the baseline prediction. An improvement on independent test observations can therefore reflect regularization without additional information about the target.

With independent measurements, ridge MSE falls further, to 0.3307. Conditional noisy copies yield no monotone result: four worsen ridge relative to the baseline, while sixteen improve it. These outcomes show how representation, nuisance predictors, sample size, and tuning interact. The simulation explains why neither feature count nor a test-loss gain identifies an information gain; it does not estimate the latent measurement precision of the observed financial characteristics.

VI. Uncertainty and External Validity

A. What the intervals measure

Numerical checks, sampling intervals, and independent confirmation answer different questions. Transport checks assess whether implementations solve the stated common problem. The paired intervals measure variation in recorded outcomes conditional on the fitted decisions. Neither addresses the selection of the research design after related work on the same data.

Predictive-loss intervals use 2,000 paired circular bootstrap draws with 12-month blocks. Library comparisons and the 13 family omissions have separate simultaneous bands over their stated comparison sets. Economic outcomes add 24- and 60-month blocks; their copy bands cover 13 interventions within each method, cost scenario, metric, and block length.

They do not jointly cover every empirical choice. Long blocks use few distinct stretches of the 179-month evaluation, and the bootstrap does not refit the learning procedures. An interval containing zero does not establish economic equivalence, which would require a prespecified margin and an interval contained within it.

B. Scales for attribution and holdings

Temporal changes provide descriptive context for attribution distance. In the original representations, mean adjacent-month half- ℓ_1 share changes range from 0.13 to 0.26 across methods; twelve-month changes range from 0.34 to 0.62. For UPSA, the adjacent-month value is 0.169, compared with 0.274 for balancing at the same date. Temporal changes combine new observations, economic conditions, and tuning; they are not a sampling-noise estimate for the representation contrast.

Balancing changes UPSA target holdings by 0.191 of average gross exposure, compared with 0.227 to 0.277 for the other methods. This is a same-date difference in named-stock targets, not realized turnover or a welfare measure. Its denominator also differs from that of covariance-share distance. High monthly return correlations in many comparisons are consistent with changed tilts within related strategies.

C. Ranking and data coverage

The rank-policy comparison rebuilds all 311 formation panels using exact numeric ties and refits all 93 linear and 60 managed-return specifications, including holdings. Only two of 16,647 linear model-month regularization choices change. Under exact ranks, original and balanced UPSA gross Sharpe ratios are 1.1261 and 1.1414, compared with 1.1262 and 1.1413 under the baseline rule. Their attribution distance changes from 0.273664 to 0.273684 and their holdings distance from 0.190589 to 0.190611.

Across 140 corresponding holdings rules, the largest absolute change in gross annual mean is 0.0317 percentage points. Local responses can be larger: the maximum single-month change at 25 basis points is 47.13 basis points, and one original single-ridge target comparison reaches 0.569 of average gross exposure. Internet Appendix Tables IA.31 and IA.32 report, respectively, differences between rank policies and representation contrasts within each policy. IPCA and boosting are outside this sensitivity analysis. These descriptive comparisons do not constitute an equivalence test.

Raw-value coverage also varies. Of 153 characteristics, 17 have finite observations for less than 80% of eligible stock-months and six for less than 50% over the 311 formation months. Two of the baseline 39 are below 80%. Neutral imputation is part of the evaluated

procedure, so coverage can affect the response to an addition. Coverage alone establishes neither accuracy nor historical availability.

D. Valuation and the scope of the evidence

Unobserved returns materially affect economic magnitudes. Original UPSA’s annual mean changes from 4.79% under the primary valuation to 8.98% when missing positions receive -100% total returns and 0.62% at $+100\%$. The strategy is net short those positions on average. These scenarios do not estimate the missing outcomes or establish that they are mostly delistings. Similar gross missing exposure across representations also cannot ensure cancellation in a paired return contrast: the signed holdings may differ.

For UPSA, the balanced-minus-original annual mean difference is -0.038 percentage points in the primary no-fee scenario, 0.020 when missing total returns are -100% , and -0.095 when they are $+100\%$. Each scenario uses its own recursively calculated ledger. These are descriptive paired contrasts; the reported bootstrap analysis does not provide intervals for the two missing-return scenarios. Their different signs show why the primary valuation cannot settle the payoff comparison.

The data are a retrospective U.S. extraction used in developing the design. Chronological estimation protects the stated return cutoff but does not correct vendor revisions or prior design selection. Stronger generalization would require a separately fixed design on another market or later observations, along with improved valuation and execution inputs.

The practical contribution is consequently a diagnostic: record how a library update changes the fitted attribution, stock targets, and net payoff under a specified implementation. Its relevance would weaken if the effects were concentrated in poorly observed characteristics or disappeared under feasible portfolio constraints. Conversely, independently confirmed net gains from actual additions would support their decision value despite sensitivity to the copy controls.

VII. Conclusion

Changing the representation of a characteristic library can change a tuned investment rule without adding observations. On a common U.S. panel, selective copying and theme balancing alter economic-theme allocations and named-stock targets across four portfolio procedures. The consequences for realized returns depend on the method, trading costs, and missing-return valuation. The experiments identify representation sensitivity, not an optimal library or a general loss from characteristic proliferation.

For a manager assessing a library update, the useful comparison follows the update through estimation, exposures, trades, and payoffs. An information-preserving control re-

veals whether part of the procedure responds to how existing inputs are represented. Actual additions still require a separate performance assessment: their small and uncertain forecasting effects here do not establish that they lack information. Independent confirmation and stronger valuation and execution inputs are needed before the measured effects support a broader investment conclusion.

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Table I
Attribution, stock holdings, and gross performance

Method	Attribution	Holdings	SR O	SR B
Blocked ridge	0.231	0.277	0.366	0.348
SR-weighted	0.198	0.239	0.490	0.397
LOO ridge	0.271	0.227	1.003	1.105
UPSA	0.274	0.191	1.126	1.141

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. Attribution is the monthly average of $\frac{1}{2} \sum_g |s_{g,B} - s_{g,O}|$, where s_g is a theme payoff's covariance contribution divided by total payoff variance. Holdings is the monthly ℓ_1 difference between B and O stock target weights divided by their average gross exposure. It is not realized turnover. SR O/B are annualized gross Sharpe ratios for original/balanced portfolios. Attribution and holdings have different units; signed covariance shares need not lie between zero and one.

Table II
Column multiplicity in the actual library

Theme	Columns	Mean PR	Count/PR
Accruals	6	3.36	1.79
Debt Issuance	7	5.14	1.36
Investment	22	5.74	3.84
Low Leverage	11	6.04	1.82
Low Risk	18	4.01	4.49
Momentum	8	2.58	3.10
Profit Growth	12	7.94	1.51
Profitability	11	4.01	2.74
Quality	17	3.47	4.90
Seasonality	12	11.78	1.02
Short-Term Reversal	6	2.40	2.50
Size	5	2.24	2.23
Value	18	4.88	3.71

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. The 153 characteristics and 13 themes follow the Jensen–Kelly–Pedersen library. For correlation eigenvalues ℓ_j , participation rank (PR) is $(\sum_j \ell_j)^2 / \sum_j \ell_j^2$. Each row averages the 120-month training windows for the 179 decisions. Count/PR averages window-specific ratios, not the ratio of displayed averages. PR describes overlap under the centered-rank encoding; it does not identify independent information sources.

Table III
Composition changes in canonical theme payoffs

Method / encoding	Value	Quality	Low risk	Distance
Blocked ridge O	0.188	0.258	0.133	0.000
Blocked ridge B	0.163	0.196	0.097	0.231
Blocked ridge V5	0.447	0.183	0.097	0.299
SR-weighted O	0.119	0.347	0.079	0.000
SR-weighted B	0.113	0.279	0.066	0.198
SR-weighted V5	0.280	0.284	0.060	0.267
LOO ridge O	0.022	0.274	0.119	0.000
LOO ridge B	0.026	0.211	0.073	0.271
LOO ridge V5	0.151	0.247	0.123	0.223
UPSA O	0.032	0.324	0.141	0.000
UPSA B	0.046	0.248	0.098	0.274
UPSA V5	0.170	0.312	0.139	0.264

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. O denotes original, B theme-balanced, and V5 five total copies of every value characteristic. Entries are average training-window covariance shares, expressed as fractions. Distance is the mean half- ℓ_1 change in all 13 shares relative to O, although only three theme shares are displayed. Shares may be negative and distance may exceed one. Before the stock gross-exposure cap, each direction with nondegenerate training volatility targets 10 percent annualized training volatility. The value-copy rows illustrate one of 13 interventions; Tables IA.6–IA.9 report all families.

Table IV
Realized stock performance under common allocation rules

Method / encoding	Gross SR	Gross mean	25 bp mean	Spread mean
Blocked ridge O	0.366	2.77	-0.63	-3.11
Blocked ridge B	0.348	2.63	-0.95	-3.56
Blocked ridge V5	0.254	2.00	-1.28	-3.65
SR-weighted O	0.490	3.66	0.01	-2.54
SR-weighted B	0.397	3.07	-0.67	-3.28
SR-weighted V5	0.370	2.91	-0.81	-3.47
LOO ridge O	1.003	4.53	0.68	-1.92
LOO ridge B	1.105	4.78	0.82	-1.85
LOO ridge V5	1.069	4.80	1.06	-1.51
UPSA O	1.126	4.79	1.10	-1.43
UPSA B	1.141	4.75	0.92	-1.70
UPSA V5	1.150	5.07	1.47	-1.03

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. O denotes original, B theme-balanced, and V5 five total copies of every value characteristic. SR is the annualized Sharpe ratio; means are annualized arithmetic excess returns in percent. A 10 percent training-volatility target is followed by a gross-exposure cap of two, so realized risks differ. Costs multiply full traded dollars and each scenario has a separate net-asset-value account. Spread uses half the spread proxy, clipped to 5–500 basis points (bp); impact and borrow fees are omitted. Missing outcomes receive zero vendor excess return. The supplementary portfolio results also report each representation under minus/plus 100 percent total-return valuation scenarios; those scenario levels are not confidence bounds.

Table V
Turnover, risk, and valuation coverage

Method	Trades	Gross	Vol. (%)	Missing	ADV (%)
Blocked ridge	1.13	1.92	7.58	1.26	97.49
SR-weighted	1.22	1.95	7.47	1.21	97.30
LOO ridge	1.28	2.00	4.52	1.29	97.62
UPSA	1.23	2.00	4.26	1.30	97.59

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. Original gross portfolios only. Trades is mean full absolute monthly trading per dollar of net asset value (NAV), including return drift and liquidations; Gross is mean absolute exposure per dollar of NAV. Vol. is annualized standard deviation in percent. Missing is mean gross exposure to unobserved future returns, as percent of NAV. Average dollar volume (ADV) coverage averages monthly traded-dollar fractions with positive ADV60 and at least 60 underlying daily observations. No participation cap is imposed; coverage below 100 percent prevents interpreting covered-trade capacity as whole-portfolio capacity.

Table VI
Actual additions and matched prediction controls

Procedure	R^2 (%)	Δ loss	95% interval
Panel A: Monthly models; reference = monthly mean			
Monthly mean	0.239	0.000	[0.000, 0.000]
Ridge 39	0.214	0.025	[0.003, 0.051]
Ridge 36 proxy	0.203	0.035	[-0.009, 0.088]
Ridge 153	0.206	0.033	[-0.017, 0.093]
Ridge balanced	0.207	0.032	[-0.018, 0.094]
Ridge Gram	0.206	0.033	[-0.008, 0.085]
Panel B: Annual models; reference = annual mean			
Annual mean	0.319	0.000	[0.000, 0.000]
Boosting 39	0.304	0.016	[-0.004, 0.041]
Boosting 153	0.326	-0.006	[-0.062, 0.040]
IPCA 36, K3,U	0.302	0.017	[-0.065, 0.113]
IPCA 153,K3,U	0.193	0.127	[-0.006, 0.275]
Panel C: Refit schedule; reference = monthly mean			
Annual mean	0.319	-0.081	[-0.193, 0.030]

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each panel names its reference. R^2 is one minus squared prediction loss divided by zero-forecast loss, in percent. Δ loss is the paired model-minus-reference change in percentage points of that same denominator; negative values favor the model. Mean forecasts use the same 120-month histories and refit schedules as the corresponding models. The intervals use 2,000 paired circular bootstrap draws with 12-month blocks, conditional on fitted decisions. Gram denotes the characteristic-Gram penalty. IPCA is instrumented principal component analysis; K3/U denotes three factors and an unrestricted intercept. The annual-mean reference row is identical to itself by construction. Panel C isolates the difference between annual and monthly refitting.

Table VII
Precision and no-information controls in a known model

Measurements	Count	Bayes MSE	Ridge MSE	Δ MSE
Exact copies	1	0.7500	0.7999	0.0000
Exact copies	4	0.7500	0.7805	-0.0194
Exact copies	16	0.7500	0.7695	-0.0304
Conditional noise	1	0.7500	0.7999	0.0000
Conditional noise	4	0.7500	0.8125	0.0126
Conditional noise	16	0.7500	0.7907	-0.0092
Independent meas.	1	0.7500	0.7999	0.0000
Independent meas.	4	0.4500	0.4856	-0.3143
Independent meas.	16	0.3088	0.3307	-0.4692

Synthetic experiment: h, e_j, ϵ are mutually independent standard Gaussian variables and $y = h + 0.5\epsilon$. Measurement noise has standard deviation one. Exact copies repeat $x_1 = h + e_1$; conditional copies retain x_1 and add $x_1 + e_j$ for $j \geq 2$; independent measurements are $h + e_j$. Count includes the retained first measurement. Each mean squared error (MSE) averages 100 replications, with 120 training, 120 validation and 4,000 test observations, plus eight nuisance predictors. Δ MSE is the paired change from count one; negative values improve prediction. Exact-copy penalty transport reproduces the count-one fit. Only independent measurements reduce the population Bayes MSE.

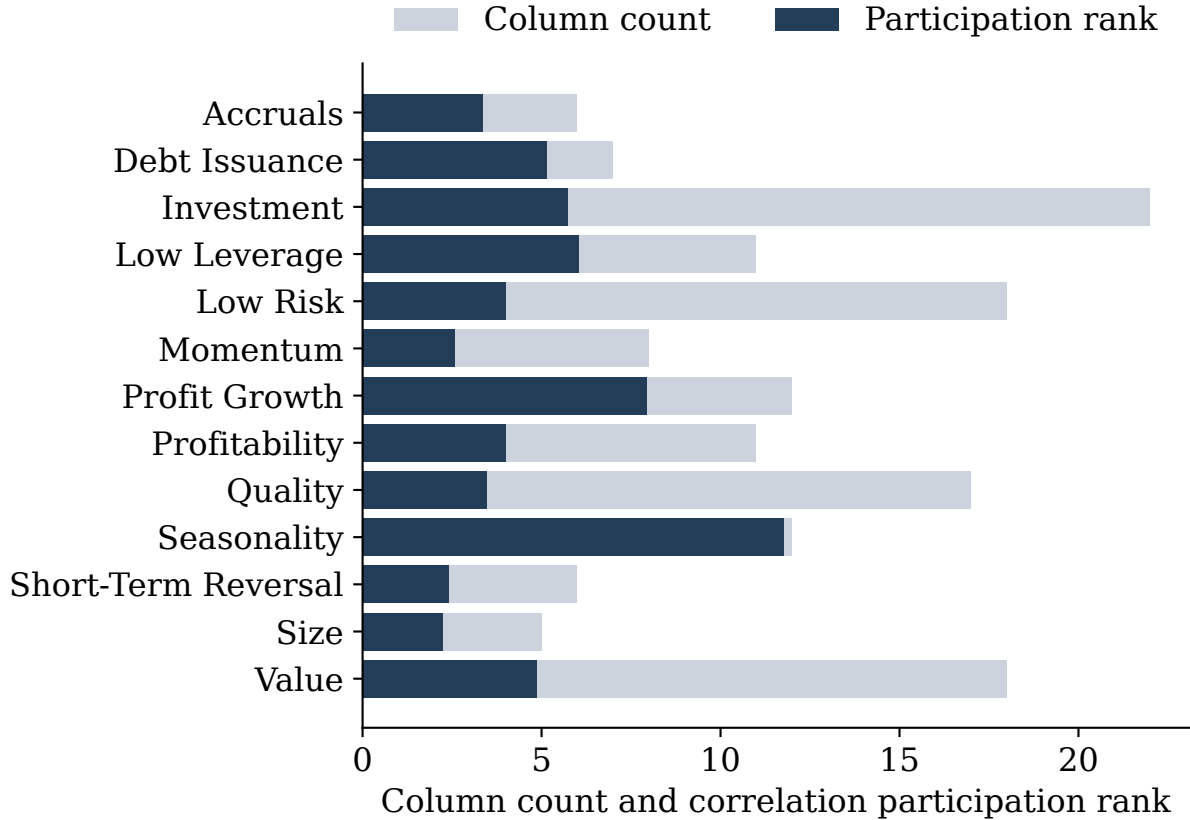


Figure 1. Uneven overlap within economic themes

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Column counts and correlation participation ranks (PR) refer to the 13 Jensen–Kelly–Pedersen themes. PR is $(\sum_j \ell_j)^2 / \sum_j \ell_j^2$, with ℓ_j the within-theme correlation eigenvalues. Ranks average 120-month training windows. The bars describe overlap, not unique predictive information or independent measurement precision.

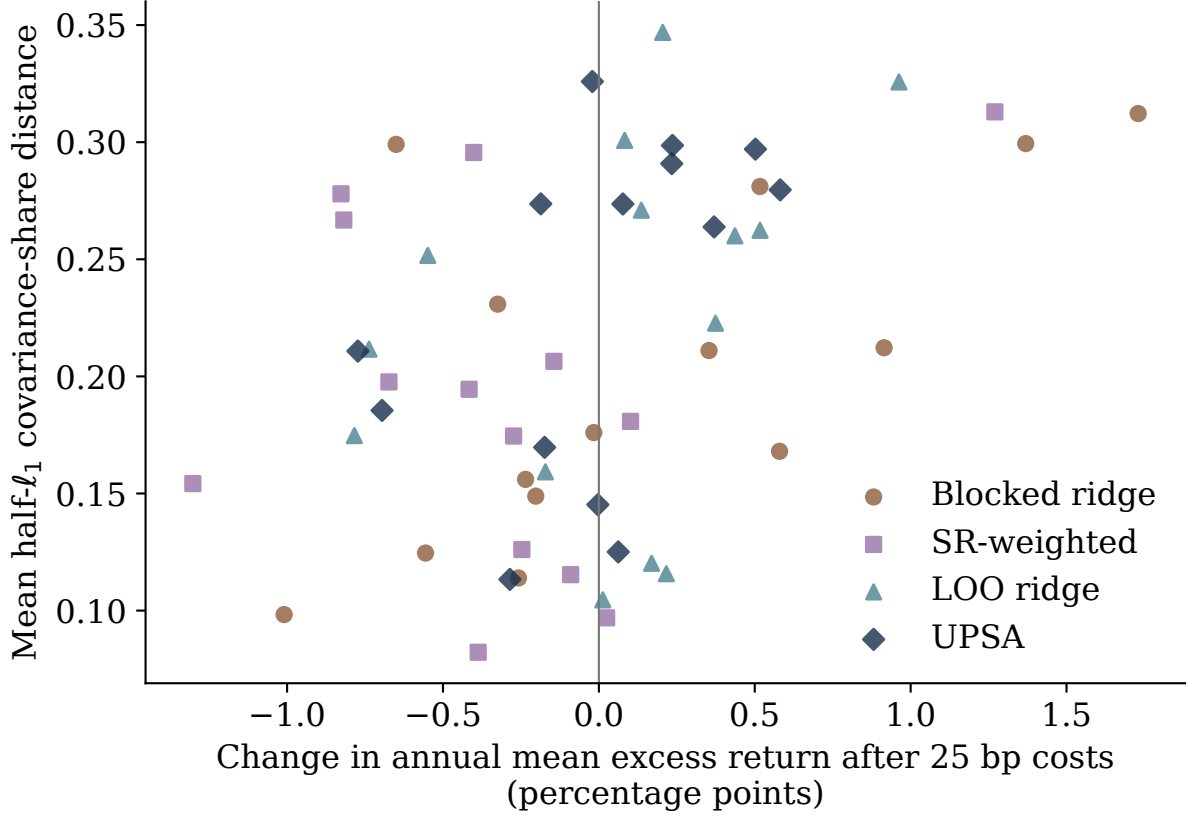


Figure 2. Changes in theme attribution and net portfolio returns

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. Each point compares one representation with the original under the same method: theme balancing or fivefold replication of one of 13 families. The horizontal axis is the change in annual mean excess return, in percentage points, after 25 basis points per traded dollar. The vertical axis is mean half- ℓ_1 distance between 13 training covariance shares. Returns include the common stock allocation and gross cap. Confidence intervals are reported separately; point proximity to zero does not establish equivalence. The plot does not estimate a causal relation between attribution distance and performance.

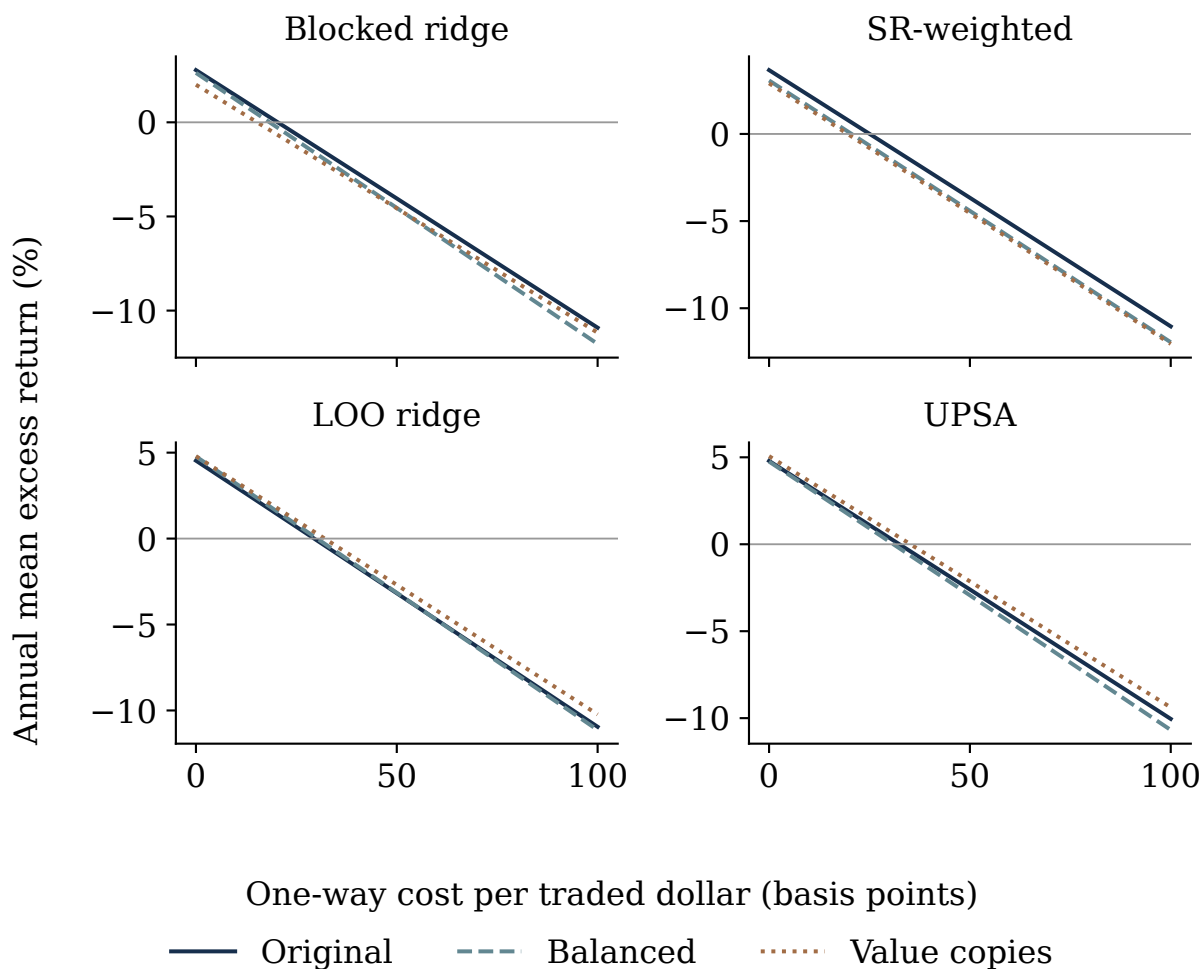


Figure 3. Cost sensitivity of a fixed allocation rule

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage. Each panel varies one-way costs per traded dollar for the original representation, theme balancing and fivefold value replication. The vertical axis is annual arithmetic mean excess return in percent. Each rate has a separate self-financing cash and return-drift account; target-weight rules remain fixed. The stock gross cap is two and primary missing outcomes receive zero vendor excess return. The curves do not include optimized trade scheduling, borrow fees or market impact. Representation-specific missing-return scenario levels are provided with the supplementary portfolio results.

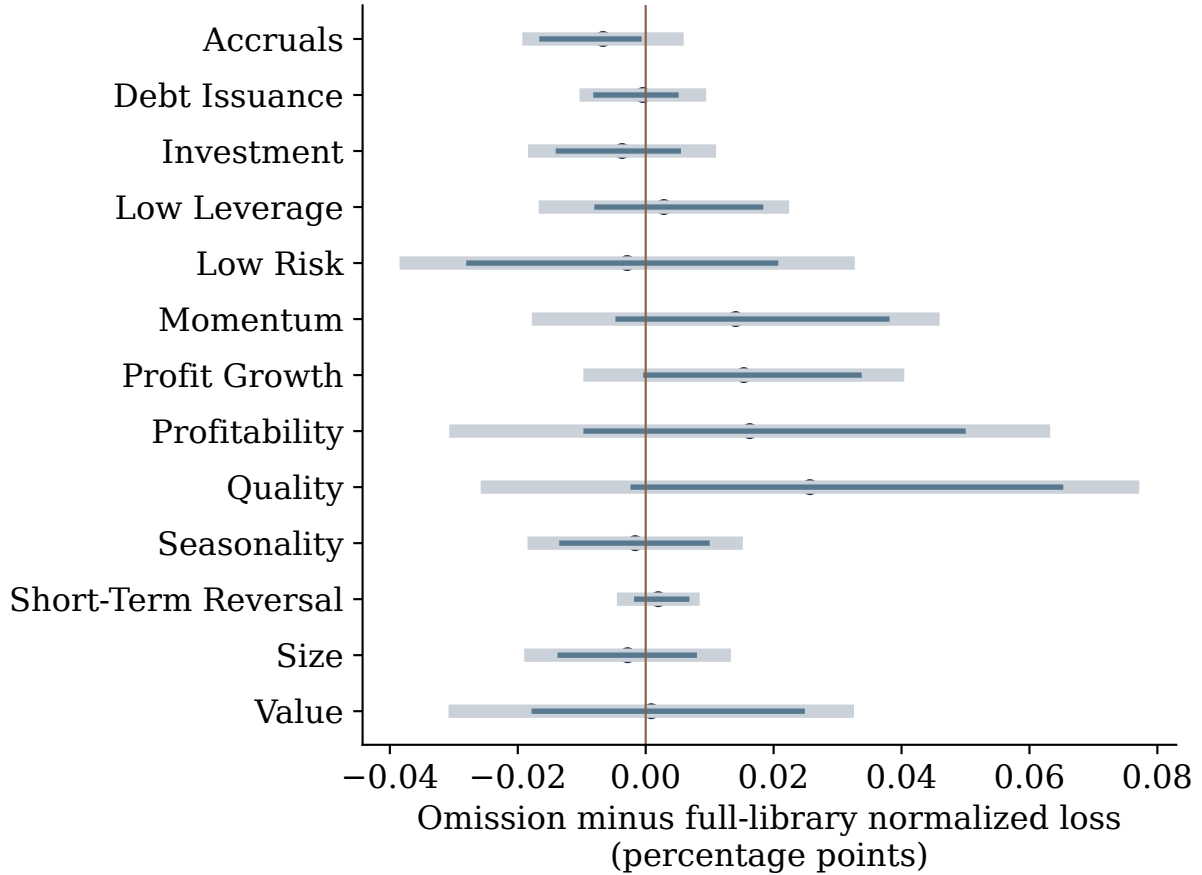


Figure 4. Family omissions on fixed stock targets

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each monthly ridge fit omits one family from the full 153-characteristic library and retunes the remaining inputs. The horizontal axis is omitted-minus-full loss, normalized by common zero-forecast loss, in percentage points. Positive values mean worse prediction. Dark intervals are pointwise 95 percent intervals; light bands are simultaneous across the 13 omissions. Both use 2,000 paired circular bootstrap draws with 12-month blocks and condition on the fitted decisions. An omission removes shared and distinct content together.

Internet Appendix to Characteristic Libraries and Portfolio Decisions

Mingyang Liu
Imperial Business School
September 2026

This appendix describes the data construction, estimation, and supporting comparisons. Section I provides the transport and attribution identities. Section II details the empirical procedures. Section III studies matched factor decompositions in IPCA. The tables report the characteristic dictionary, all library interventions, inference, rank-policy sensitivity, and fixed-asset diagnostics.

I. Transport and Attribution Identities

A. The effective penalty of a redundant representation

Let $A \in \mathbb{R}^{p \times q}$ have full row rank and let $b = A\theta$ denote penalized slopes. The minimum-norm representative is $\theta_* = A'(AA')^{-1}b$. Every other representative has the form $\theta_* + v$ with $Av = 0$. Since θ_* is in the row space of A , it is orthogonal to v . Hence

$$\|\theta_* + v\|^2 = b'(AA')^{-1}b + \|v\|^2. \quad (\text{IA.1})$$

Minimizing over redundant coordinates gives the effective native penalty stated in the text. Exact selective copies make AA' diagonal with entries equal to copy counts, yielding a penalty b_j^2/m_j for a coordinate with m_j copies. Under uniform copies the effective penalty is proportional to the original identity, so multiplying the scalar penalty by the copy count restores the problem.

For a transported canonical penalty, both the loss and $\theta'A'PA\theta$ depend only on $A\theta$. Surjectivity of A onto the canonical slope space makes their attainable objective values identical to those of the canonical problem. If that problem has a unique fitted coefficient vector, every expanded minimizer maps to it. Unique redundant coefficients are unnecessary. With an unpenalized intercept, apply this argument after profiling it or transport its null penalty block explicitly. Arbitrary changes mixing that null subspace with penalized slopes require the corresponding full transformation.

B. Variance-game Shapley allocation

Fix fitted component payoffs with finite second moments. For the game $v(S) = \text{Var}(\sum_{g \in S} u_g)$, the marginal contribution of component g to a preceding coalition is

$$v(S \cup \{g\}) - v(S) = \text{Var}(u_g) + 2 \sum_{h \in S} \text{Cov}(u_g, u_h). \quad (\text{IA.2})$$

In a uniformly random permutation each other component precedes g with probability one half. The Shapley value (Shapley, 1952) is therefore $\text{Cov}(u_g, \sum_h u_h)$. Coefficients, component definitions, and the covariance sample must remain fixed in this argument. Refitting a model for each coalition defines another game. Different games assign different meanings to the contribution of a theme, so their allocations need not coincide (Sundararajan and Najmi, 2020).

For our distinct projection game, let C be the covariance matrix of scalar theme payoffs, $c = C\mathbf{1}$, and $v = \mathbf{1}'C\mathbf{1} > 0$. A coalition's explained fraction is $c'_S C_{SS}^+ c_S / v$. The implementation enumerates all 2^{13} coalitions using a pseudoinverse with relative threshold 10^{-10}

and averages marginal increments with exact Shapley combinatorial weights. “Exact” here means full enumeration for that numerical game; it does not mean population identification or invariance to taxonomy. Zero target variance is handled as undefined attribution under the declared cash convention.

II. Data Construction and Estimation

A. Source provenance and the estimation calendar

The reconstruction uses 26 annual United States files from the archived JKP characteristics, covering 2000 to 2025. The 153 characteristic names and their 13 theme assignments come from the archived classification associated with Jensen, Kelly, and Pedersen (2023). We preserve these inherited cluster labels throughout the interventions. For example, the source places `seas_16_20na` in Accruals and `capex_abn` in Debt Issuance; membership is not reassigned from variable names. It is a retrospective taxonomy, not a sequence of classifications discovered by investors in real time. Source-file hashes identify the vendor vintage used by the calculation; they do not establish historical publication lags or point-in-time availability.

Formation dates and return dates are separate indices. A January formation observation predicts the following February return. The estimation calendar retains formation months with at least 200 finite forward returns: January 2000 to November 2025, or 311 months. The evaluation window contains 179 formation months, January 2011 to November 2025, with realizations through December 2025. December 2025 raw characteristics are available but fall outside this estimation calendar. The source archive is retrospective, so chronological estimation does not establish point-in-time availability or an untouched confirmation sample.

B. Formation screens, ranks, and raw characteristic coverage

A security is eligible when the four source indicators for common stock, main exchange, primary security, and main observation equal one; its size group is small, large, or mega; and its market equity is finite and positive. Missing indicators fail the screen. The source identifier and month jointly identify observations. No forward-return requirement enters security eligibility or characteristic ranking.

For characteristic j , let n_{jt} be its finite count among the N_t eligible securities. A finite value receives its average rank among finite observations, divided by $n_{jt} + 1$, minus one-half. Tied values share the average rank. The baseline groups adjacent sorted finite values whose differences are no greater than 10^{-13} . This explicit rule reproduces the nullable-column ranking behavior of pandas 2.2.2. The exact-rank sensitivity uses zero tolerance with separate panel construction and re-estimation. Nonfinite values receive zero after this

transformation. Thus a transformed zero can denote a neutral finite observation, a tie, or an imputation. Its frequency is not a missing-data statistic. A constant characteristic produces no cross-sectional direction. The outcome mask is applied only after all formation ranks have been calculated.

Characteristic availability is measured from finite raw values after applying the estimation-universe screen. The panel contains 700,624 eligible security-months. Six characteristics have pooled finite coverage below 50% and 17 below 80%. Two of the baseline 39 fall below 80%: `dsale_dinv` and `dsale_dsga`, at 70.64 and 77.01%. Annual and pooled counts use explicit denominators; pooled coverage weights security-months, whereas the separately reported monthly average weights months equally. Availability differences are documented without attributing model-performance differences to missingness alone.

C. Moment objectives and the unpenalized intercept

Write $u_{it} = (1, z'_{it})'$ for the constant and ranked characteristics, and let $\mathcal{O}_t$ contain securities with finite forward excess returns $y_{i,t+1}$. Define $n_t = |\mathcal{O}_t|$ and

$$G_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} u_{it} u'_{it}, \quad x_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} u_{it} y_{i,t+1}, \quad q_t = \frac{1}{n_t} \sum_{i \in \mathcal{O}_t} y_{i,t+1}^2. \quad (\text{IA.3})$$

The distinct ex-ante augmented moment is $\mathcal{Q}_t = N_t^{-1} \sum_{i=1}^{N_t} u_{it} u'_{it}$. For coefficient β , equal-month prediction loss is $\bar{q} - 2\bar{x}'\beta + \beta' \bar{G} \beta$. These moments exactly reconstruct observed-stock squared error; $\mathcal{Q}_t$ does not substitute for G_t in that objective.

Partition $\beta = (a, b)'$. Profiling the unpenalized intercept gives $a = \bar{x}_0 - \bar{G}_{0z} b$, centered slope moment $C = \bar{G}_{zz} - \bar{G}_{z0} \bar{G}_{0z}$, and score $s = \bar{x}_z - \bar{G}_{z0} \bar{x}_0$. Ridge solves $(C + \lambda P)b = s$. Conventional precision is $P = cI$, with $c = \text{tr}(\bar{G}_{zz})/p$ for the selected p slopes. The theme rule multiplies its diagonal by $13n_f/p$ for a characteristic in theme f of size n_f ; the metric rule uses $\bar{Q}_{zz}$. These are different penalties, not interchangeable information measures. An additional exact mean candidate sets $b = 0$. The moment formulas and direct stock regressions give the same objective, including when feature means are nonzero.

D. Nested estimation and decision timing

For a monthly-refitted decision at formation t , the outer history consists of formation months $t - 121, \dots, t - 2$. Their forward outcomes end at $t - 1$. Formation $t - 1$ is excluded from estimation. Within the 120-month history, the first 95 months fit candidates, the next month is omitted, and the final 24 months select a candidate. Stock ridge minimizes equal-month validation squared error over $\lambda \in \{.001, .01, .1, 1, 10, 100, 1000, 10000\}$ and the exact mean. The selected candidate is refitted on all 120 outer months, including the internal gap.

For January 2011, the outer formation history is December 2000 to November 2010. The corresponding forward outcomes end in December 2010. Each precision depending on training moments is recalculated separately on the inner and full outer sample. Managed blocked methods use the same temporal split but their own portfolio objective and grid. IPCA and boosting update model parameters each January and hold them fixed for the remainder of the calendar year. All evaluation losses compare common dates and observed-return stocks. This timing controls the implemented information cutoff, conditional on the retrospective source values.

E. Duplication, penalty transport, and information controls

Let Z contain the canonical slopes and encode them as ZA , where A has full row rank. Encoded coefficients θ imply $b = A\theta$. The minimum Euclidean penalty among representatives of b is

$$\min_{A\theta=b} \theta'\theta = b'(AA')^{-1}b. \quad (\text{IA.4})$$

Consequently, giving a selected coordinate m total copies divides its induced precision by m . The 39 selective interventions apply two, five, or ten copies to each of 13 baseline themes, retaining the canonical scalar normalization. A separate counterfactual fixes the baseline's selected penalty before allowing retuning. Squared total prediction change equals the two squared components plus twice their inner product; component magnitudes are not an additive decomposition of squared distance.

Transport instead penalizes $\theta'A'PA\theta$. Loss and penalty then depend only on $A\theta$, preserving the canonical optimum when unique even though redundant coefficients are not unique. The intercept is outside these slope transformations. The 117 explicit expanded stock-ridge fits compare native-coordinate estimation with its induced canonical penalty. Separate uniform-copy and synthetic transport controls assess unchanged-rule identities; recodings with condition numbers up to 100 assess finite-precision sensitivity. These checks concern the specified quadratic subproblems, rather than a transported implementation of every managed-return procedure. The underlying algebra is consistent with the companion metric analysis (Liu, 2026b,a).

The synthetic experiment separately generates $x_j = h + \tau e_j$ with mutually independent standard Gaussian h, e_j , and noisy transformations of a retained x_1 . Independent measurements reduce latent Bayes variance to $\tau^2/(\tau^2 + m)$; conditional noisy copies add no information about h given x_1 . They can still alter finite-sample ridge behavior. The experiment uses 100 replications per noise level, separate training, validation, and test streams, and an independently solved transport check. It does not estimate measurement precision for the real characteristic library.

F. Managed returns and shrinkage conventions

Characteristic j generates the managed payoff $r_{jt} = N_t^{-1} \sum_i (z_{ijt} - \bar{z}_{jt}) \tilde{y}_{i,t+1}$, where $\tilde{y}$ sets missing vendor excess returns to zero for the stated primary valuation. This uses ex-ante holdings and denominator. In moment notation it is $(n_t/N_t)(x_{jt} - \bar{z}_{jt}x_{0t})$. Direct stock reconstruction gives the same managed payoffs in all 311 months. The managed experiment combines four procedures with 15 representations: original, positive theme scaling, and five total copies of each theme separately.

For a $T \times p$ training matrix R of managed payoffs, the regression-on-one objective has no fitted intercept. It uses $M = R'R/T$ and $\mu = R'\mathbf{1}/T$, not centered covariance. Encoded returns are $\tilde{R} = RA$. Coordinate ridge solves $(\tilde{R}'\tilde{R}/T + zI)^{-1}\tilde{R}'\mathbf{1}/T$. The common grid is $10^{-7}, \dots, 10^3$. Blocked validation selects the largest average $v - v^2/2$. The Sharpe-weighted adaptation replaces I by a diagonal vector proportional to $1/\max(\widehat{\text{SR}}_j^2, 10^{-6})$, using monthly sample variance with denominator $T - 1$ and normalizing the diagonal vector to Euclidean norm one. It recalculates this vector on each training sample. These conventions define the adaptation used here, which differs from an exact replication of Ke and Pourmohammadi (2025).

The authors' UPSA implementation (Kelly et al., 2026; Pourmohammadi, 2026) computes, for every grid value,

$$H_z = \tilde{R}(\tilde{R}'\tilde{R}/T + zI)^{-1}\tilde{R}'/T, \quad V_{iz} = \frac{(H_z\mathbf{1})_i - H_{z,ii}}{1 - H_{z,ii}}. \quad (\text{IA.5})$$

This deleted-row convention retains penalty Tz in the unnormalized normal equations. It is historical leave-one-out selection, not sequential trading within that history. UPSA chooses $a \geq 0$ minimizing $a'V'Va/(2T) - \bar{V}'a$, with no sum-to-one constraint. Asset weights may be signed. Its ridge comparator selects the largest $\bar{V}_z/\sqrt{\bar{V}_z^2}$. For $f(d) = \sum_z a_z/(d + z)$, author trace scaling is $\sum_d 1/f(d) / \sum_d d$, using the eigenvalues stored by the primal or dual implementation. Author trace normalization precedes the common risk overlay, and the replication data retain coefficients at both stages. Actual training windows have full row rank. The implementation requires full row rank for wide inputs; this condition holds in the empirical training windows.

G. Annual IPCA estimation and matched sections

The IPCA comparison follows the characteristic-dependent factor representation of Kelly, Pruitt, and Su (2019). It crosses the historical 36-characteristic proxy and full 153-characteristic panel with one, three, or five latent factors and restricted or unrestricted intercept structure, yielding 12 models. A constant characteristic is included; observed

factors are absent. January estimation uses 120 training months with equal month weights, a maximum of 500 alternating iterations, and tolerance 10^{-6} . All 180 recorded annual fits met that convergence criterion. The stopping criterion does not establish global optimality.

For estimated intercept direction α , loadings Γ , and training factor mean $\bar{f}$, the augmented forecast coefficient is $\beta = \alpha + \Gamma\bar{f}$. These quantities remain fixed until the next January. The metric section uses the outer-training ex-ante moment Q and

$$d = (\Gamma'Q\Gamma)^{-1}\Gamma'Q\alpha, \quad \alpha_Q = \alpha - \Gamma d, \quad \bar{f}_Q = \bar{f} + d. \quad (\text{IA.6})$$

Hence $\alpha_Q + \Gamma\bar{f}_Q = \beta$. The same shift is applied to factors extracted from evaluation returns when reporting contemporaneous fitted-return loss. Those realized factors are not used for next-return forecasting. Complete forecasts and fits are checked for equality across sections. A factor-only forecast can change because it omits the correspondingly shifted intercept; it is therefore a section-dependent descriptive object, not a distinct improvement in the complete forecasting rule.

H. Annual boosting and the mean candidate

Histogram gradient boosting provides a nonlinear prediction comparison for the baseline 39 and full 153 characteristics. Inputs use the same formation ranks and neutral imputation as the linear models. Candidate trees have seven or 15 maximum leaves, 50 or 100 iterations, learning rate .05, minimum leaf size 50, and L_2 regularization 10. Early stopping is disabled. The random seed is fixed. This specified small-grid benchmark evaluates a limited subset of the nonlinear designs discussed by Gu, Kelly, and Xiu (2020).

Within each fitting or validation block, observed-return stock weights are proportional to the reciprocal of that month's observed count and are normalized to mean one. This preserves equal month contributions while fixing the software's regularization scale. January selection compares the four tree configurations with the inner-sample historical mean. The chosen procedure is refitted on the full outer history, then predicts every eligible stock during that year. Missing outcomes remain represented in saved predictions but are omitted from supervised loss.

The mean is selected in 25 of 30 annual decisions: 12 of 15 for the baseline and 13 of 15 for the full library. The annual-mean comparison is exploratory: it was selected after examining boosting performance and was not prespecified. It isolates the refit schedule by keeping the mean fixed between January updates. These outcomes concern the stated grid and timing. Boosting is evaluated for stock prediction and does not enter the common holdings calculation.

I. The distinct attribution games

Attribution uses fixed original theme names after mapping encoded estimates into canonical coefficients. For forecast components, the covariance is induced by the centered training ex-ante characteristic moment. For managed components g_f , it is the sample time-series covariance of the 120 training payoffs. These are different populations of variation. In either case, with $g = \sum_f g_f$, the covariance contribution $\phi_f = \text{Cov}(g_f, g)$ sums to $\text{Var}(g)$. It also equals the Shapley value of the specific game $v(S) = \text{Var}(\sum_{f \in S} g_f)$. Shares can be negative; they are undefined at zero total variance. Their half- L_1 distance is not total variation between probabilities and can exceed one.

The sensitivity game projects the fixed total managed payoff on the *scalar theme component payoffs* in a coalition, without refitting the return model. If C is their covariance and $c = C\mathbf{1}$, its normalized value is $c'_S C_{SS}^+ c_S / (\mathbf{1}' C \mathbf{1})$. All 8,192 coalitions are enumerated with pseudoinverse relative cutoff 10^{-10} for eight original/balanced procedures at three selected dates. This is not projection onto every underlying characteristic or nonlinear conditional-expectation attribution. Shapley (1952) supplies the allocation principle; Sundararajan and Najmi (2020) explains why the game must be specified. Neither statistic measures a refitted predictive-loss contribution, marginal net utility, or a priced risk.

J. Stock holdings, financing, and vendor-consistent drift

A canonical slope direction forms preliminary weights $(z_{it} - \bar{z}_t)' b_t / N_t$. Its historical managed-return volatility sets a 10% annual target, followed by uniform scaling whenever current gross exposure exceeds two. Directions with annualized training volatility at or below 10^{-12} hold cash. The IPCA risk overlay updates monthly despite annual coefficient fits. Because the gross multiplier varies over time, the capped series is not a constant rescaling of the preliminary series; neither equal realized risk nor unchanged Sharpe follows.

Each fee scenario has its own NAV. Targets are dollars per pre-trade NAV. Trades compare new targets with drifted old positions on the union of old and new identifiers, including liquidations. Let κ_t be the fee fraction, r_{t+1}^F the French financing rate, r_{t+1}^V the vendor rate, and $y_{i,t+1}$ vendor excess returns. For a dollar-neutral target,

$$A_{t+1} = 1 + r_{t+1}^F + w_t' y_{t+1} - \kappa_t (1 + r_{t+1}^F), \quad w_{i,t+1}^- = \frac{w_{it} (1 + r_{t+1}^V + y_{i,t+1})}{A_{t+1}}. \quad (\text{IA.7})$$

The source rate is recovered from the same-month total-minus-excess return difference, which is common across available securities. Matching security identifiers to next-month observations verifies 402,956 of 403,129 finite forward outcomes across all 179 evaluation months, with maximum discrepancy 2.5×10^{-16} . The remaining 173 finite targets have no finite

direct next-month match in this join and are not independently verified by that check. They are distinct from the 2,710 missing forward outcomes retained in the valuation scenarios. The financing-rate difference cancels from the new neutral book’s aggregate payoff, but not from individual position drift. Fees affect subsequent NAV and trades; each cost ledger is recalculated recursively.

Turnover is the full absolute traded amount, with no half-turnover factor. Flat one-way scenarios charge 10, 25, 50, or 100 basis points. The spread scenario uses half the formation proxy, clipped to 5–500 basis points, with explicit median/fallback and stale-exit conventions. Missing future outcomes retain holdings: primary valuation sets vendor excess return to zero, while separate no-fee ledgers set missing total returns to minus or plus 100%. These are scenarios, not universal bounds. ADV coverage is the share of all traded dollars with usable ADV; capacity is calculated only across covered trades. Costs are not optimized, and the exercise is not the implementable frontier of Jensen et al. (2026).

K. Paired inference and fixed-asset diagnostics

Prediction intervals resample paired monthly loss differences using 2,000 circular block draws of length 12. The normalization by average squared outcomes is held fixed. Simultaneous bands use the bootstrap maximum absolute standardized deviation over the declared comparison family; omission comparisons form a separate 13-theme family. Economic contrasts use paired 12-, 24-, and 60-month blocks and recompute annual mean, Sharpe, and the quadratic certainty-equivalent approximation with risk aversion three. Their copy bands cover 13 interventions within each method, scenario, metric, and block length, not all empirical choices jointly.

Pairing preserves common realized months and dependence across compared procedures. It does not cancel bias from missing outcomes, data vintages, or estimated strategy selection. These intervals condition on the fitted decision sequences: they do not refit the learning algorithms in every bootstrap draw or account for prior research search. An interval containing zero does not establish economic equivalence.

The 75 fixed French test assets use realization-month excess returns. Intercept regressions include a strategy alone or the six French factors with a strategy. Standard errors use a 12-lag Bartlett Newey–West calculation and normal reference probabilities; Holm adjustment applies across 75 intercepts within each specification. The diagnostic is conditional on its fixed payoff set and generated strategy. The importance of the chosen payoff span for stochastic-discount-factor assessment is discussed by Hansen and Jagannathan (1991, 1997).

The six-factor benchmark has a root-mean-square monthly alpha of about 14.95 basis points. Adding original UPSA gives 15.07 basis points; adding balanced UPSA gives 15.35.

Strategy-only regressions leave larger intercepts, as a dollar-neutral characteristic portfolio does not replace the broad market factor. The fixed assets keep the pricing targets unchanged across library interventions, although their underlying stocks overlap with the estimation universe. These comparisons assess the specified time-series benchmark span, not equilibrium prices of risk for individual themes.

L. Data and replication materials

The replication package documents the source vintage, input fields, estimation settings, software dependencies, random seeds, and execution order from acquired inputs to exhibits. Licensed data are accessed separately from the distributed code. Synthetic test inputs demonstrate file formats and functionality; they do not reproduce the empirical estimates. The historical source vintage and development-sample design remain qualifications of those estimates.

III. Factor Attribution and the Complete Stock Index

A. A matched section in an unrestricted factor model

The companion theory develops a matched factor decomposition that preserves the complete fitted index (Liu, 2026b). We use its unrestricted-IPCA specialization as an attribution control. Following Kelly, Pruitt, and Su (2019), let $U_t = (\mathbf{1}, Z_t)$ include the constant and p slopes, let $\alpha \in \mathbb{R}^{p+1}$ be the characteristic intercept, $\Gamma \in \mathbb{R}^{(p+1) \times K}$ the loading matrix, and $f_{t+1} \in \mathbb{R}^K$ the factors. Write

$$r_{t+1} = U_t \alpha + U_t \Gamma f_{t+1} + \varepsilon_{t+1}. \quad (\text{IA.8})$$

For a positive metric Q on this augmented space with nonsingular $\Gamma' Q \Gamma$, define

$$d_Q = (\Gamma' Q \Gamma)^{-1} \Gamma' Q \alpha, \quad \alpha_Q = \alpha - \Gamma d_Q, \quad f_{Q,t+1} = f_{t+1} + d_Q. \quad (\text{IA.9})$$

Then $\Gamma' Q \alpha_Q = 0$ and

$$\alpha_Q + \Gamma f_{Q,t+1} = \alpha + \Gamma f_{t+1}. \quad (\text{IA.10})$$

The same equality holds for a forecast based on the training factor mean, provided both the mean and the residual intercept are transformed. A further invertible rotation or normalization of Γ and f_Q leaves the complete index unchanged when matched correctly.

The factor-only mean $\Gamma \bar{f}$ changes by Γd_Q , while the complete mean $\alpha + \Gamma \bar{f}$ remains fixed. This identity applies within a fitted model with a nonsingular fitted factor metric. Separately estimated libraries and changes in a penalty can change the complete forecast. A

structural interpretation of the decomposition requires additional assumptions, as developed in Liu (2026b); it does not follow from the section identity.

The distinction also limits which asset-pricing statements follow. If a researcher regresses a fixed asset return on a factor series shifted by a constant, the estimated regression intercept changes even though the span of the regressors with an intercept is unchanged. If the shift varies across annual refits, a single pooled affine argument is insufficient. We therefore verify complete stock-index equality directly month by month. The fixed-test-asset analysis uses returns on the constructed stock portfolios and reports conditional time-series intercepts. Those diagnostics evaluate a specified benchmark span rather than identify equilibrium risk prices for the section-normalized factors.

B. Evidence from fitted models

A factor-only assessment can change under a matched reallocation of factor premia and the residual characteristic intercept, even when the complete stock forecast is unchanged. We quantify this distinction in rolling IPCA fits. The section identity from Liu (2026b), stated above, supplies a within-fit attribution control. Selective duplication in penalized regression can instead change the complete fitted rule when its penalty is not transported.

We estimate restricted and unrestricted IPCA models with one, three, and five latent factors on the historical 36-name proxy and the full 153-characteristic library. All models include an explicit constant characteristic and no observed factors. There are 15 January refits per specification, each using the available 120-month training window, giving 180 fitted models. All meet the declared convergence tolerance of 10^{-6} before the 500-iteration limit. This is evidence that the implemented stopping rule was reached, not a proof that an alternating algorithm found the global optimum. The convergence tables below record iterations and final tolerances by fit.

For each unrestricted fit, we form the Euclidean-section output and the matched section under the training ex-ante Gram matrix. We shift the residual intercept and factor mean together as in equation (IA.9). We then reconstruct stock forecasts in the original named characteristic coordinates and evaluate them on the common observed-target set. Complete forecasts agree to numerical precision. The largest complete-loss discrepancy across the unrestricted specifications is approximately 1.4×10^{-17} in squared decimal-return units. Direct stock-level reconstruction gives the same agreement.

Table IA.33 reports the factor-only and complete statistics for the unrestricted models. For the full library with three factors, the factor-only Euclidean forecast has a zero-forecast R^2 of -0.246 percent, while its Gram-section counterpart has 0.224 percent. The complete forecast has $R^2 = 0.193$ percent under both sections. The sign reversal in the factor-only

statistic accompanies unchanged complete stock predictions. It therefore measures attribution sensitivity within the fitted model.

The historical 36-name proxy gives another useful comparison. With three factors, its factor-only Euclidean and Gram statistics are -0.018 and 0.267 percent, while its complete forecast reaches 0.302 percent. In this specification, including the residual characteristic intercept improves the complete forecast relative to the reported factor-only components. For the full library, the complete forecast can be worse than the Gram factor-only component. These patterns are compatible with the identity because the two components can help or hurt out of sample. There is no theorem that adding an estimated residual intercept improves future prediction.

Figure IA.1 displays the three factor dimensions separately for each library. The matched-section gap is not constant across dimensions. More importantly, the complete forecasts in the two panels differ because the models were separately estimated on genuinely different characteristic libraries. Only the within-fit matched transformation is an exact control; the cross-library difference remains an empirical comparison. An estimated complete-forecast loss can therefore vary with library size without violating the identity.

Realized-factor fit is much larger than predictive performance. For the full library, restricted-model realized-fit R^2 rises from about 19.50 percent with one factor to 23.84 percent with five factors. Its forecast R^2 is about 0.243 percent with one factor and 0.219 percent with five. The realized factor is extracted from the return vector being fitted. It can summarize contemporaneous covariance effectively without providing an equally informative estimate of the next return. Contemporaneous fit and next-return prediction therefore measure different aspects of the model.

The restricted models supply a zero-intercept control. Their residual characteristic intercept is fixed at zero, so the section shift in equation (IA.9) is zero. Factor-only and complete forecasts therefore coincide under the reported transformation. The equality follows from the zero-intercept restriction. A pricing-error interpretation of an unrestricted residual component requires a specified economic benchmark.

The complete-index comparison separates a change in component attribution from a change in the stock forecast. It supplies a reference for the library interventions: their economic outcomes depend on the complete estimated rule and the positions it generates.

The section is determined by the training characteristic metric, independently of alphas on the 75 French portfolios. The fixed-asset diagnostics use returns on the constructed stock portfolios in a stated time-series regression to assess their contribution to a fixed benchmark span. They do not identify an equilibrium premium from a section-dependent intercept.

C. Complete-index section identity

If $\Gamma'Q\Gamma$ is invertible, substitution of the definitions in equation (IA.9) gives $\Gamma'Q\alpha_Q = 0$ and

$$\alpha_Q + \Gamma f_{Q,t+1} = \alpha - \Gamma d_Q + \Gamma(f_{t+1} + d_Q) = \alpha + \Gamma f_{t+1}. \quad (\text{IA.11})$$

Averaging the factor series over the same training dates gives the corresponding complete-forecast identity. The proof does not involve future targets and applies to any common stock matrix used to evaluate the complete index. It makes no statement about separately re-estimated libraries or penalized objectives whose penalties have not been transported.

Table IA.1: Characteristic dictionary and availability

Coverage is the fraction of finite source values among 700,624 eligible stock-months from January 2000 through November 2025. Worst is the minimum annual fraction and Year identifies that year. B39 marks the fixed 39-characteristic baseline. Names and inherited cluster labels follow the Jensen–Kelly–Pedersen (JKP) library. Cluster membership is taken from the source classification, not inferred from characteristic names. Theme codes: ACCR = Accruals, DEBT = Debt Issuance, INV = Investment, LEV = Low Leverage, LRISK = Low Risk, MOM = Momentum, GROW = Profit Growth, PROF = Profitability, QUAL = Quality, SEAS = Seasonality, REV = Short-Term Reversal, SIZE = Size, VAL = Value.

Characteristic	Theme	B39	Finite (%)	Worst (%)	Year
cowc_gr1a	ACCR	Yes	89.6	85.9	2022
oaccruals_at	ACCR	Yes	96.9	90.3	2003
oaccruals_ni	ACCR	Yes	96.9	90.3	2003
seas_16_20na	ACCR	–	39.4	24.2	2000
taccruals_at	ACCR	–	96.6	90.1	2003
taccruals_ni	ACCR	–	96.5	90.1	2003
capex_abn	DEBT	Yes	83.7	68.5	2000
debt_gr3	DEBT	Yes	82.4	71.0	2000
fnl_gr1a	DEBT	Yes	98.6	94.6	2021
ncol_gr1a	DEBT	–	89.7	86.0	2022
nfna_gr1a	DEBT	–	98.6	94.6	2021
ni_ar1	DEBT	–	82.4	62.7	2000
noa_at	DEBT	–	89.5	85.7	2022
aliq_at	INV	Yes	90.9	86.7	2022
at_gr1	INV	Yes	98.6	94.6	2021
be_gr1a	INV	Yes	93.1	84.4	2021
capx_gr1	INV	–	92.8	85.1	2002
capx_gr2	INV	–	89.3	74.9	2000
capx_gr3	INV	–	85.2	70.1	2000
coa_gr1a	INV	–	90.8	86.7	2022
col_gr1a	INV	–	90.1	86.3	2022
emp_gr1	INV	–	87.5	75.6	2000
inv_gr1	INV	–	72.3	63.3	2022
inv_gr1a	INV	–	96.8	92.7	2022
lnoa_gr1a	INV	–	92.2	75.3	2000
mispricing_mgmt	INV	–	98.5	94.6	2021
ncoa_gr1a	INV	–	90.8	86.7	2022
nncoa_gr1a	INV	–	89.2	85.5	2022
noa_gr1a	INV	–	89.1	85.5	2022
ppeinv_gr1a	INV	–	87.4	83.1	2022
ret_60_12	INV	–	78.3	60.9	2000
sale_gr1	INV	–	96.6	90.3	2022
sale_gr3	INV	–	89.7	78.9	2000

Table IA.1: Characteristic dictionary and availability (continued)

Characteristic	Theme	B39	Finite (%)	Worst (%)	Year
saleq_gr1	INV	–	96.4	90.2	2021
seas_2_5na	INV	–	78.2	61.0	2000
age	LEV	Yes	100.0	100.0	2000
aliq_mat	LEV	Yes	84.4	75.2	2021
at_be	LEV	Yes	96.1	91.3	2021
bidaskhl_21d	LEV	–	99.4	99.0	2000
cash_at	LEV	–	99.4	97.5	2022
netdebt_me	LEV	–	99.4	97.5	2022
ni_ivol	LEV	–	82.4	62.7	2000
rd5_at	LEV	–	41.7	34.2	2000
rd_sale	LEV	–	49.7	46.8	2008
tangibility	LEV	–	88.1	85.3	2022
z_score	LEV	–	88.3	83.8	2021
beta_60m	LRISK	Yes	86.4	74.4	2000
beta_dimson_21d	LRISK	Yes	99.0	98.3	2021
betabab_1260d	LRISK	Yes	86.0	74.1	2000
betadown_252d	LRISK	–	97.1	92.2	2021
earnings_variability	LRISK	–	82.5	63.1	2000
ivol_capm_21d	LRISK	–	99.0	98.3	2021
ivol_capm_252d	LRISK	–	97.4	93.4	2021
ivol_ff3_21d	LRISK	–	99.0	98.3	2021
ivol_hxz4_21d	LRISK	–	99.0	98.3	2021
ocfq_saleq_std	LRISK	–	89.7	77.8	2000
rmax1_21d	LRISK	–	99.0	98.3	2021
rmax5_21d	LRISK	–	99.0	98.3	2021
rvol_21d	LRISK	–	99.0	98.3	2021
seas_6_10na	LRISK	–	61.9	41.4	2000
turnover_126d	LRISK	–	98.8	96.9	2021
zero_trades_126d	LRISK	–	98.8	96.9	2021
zero_trades_21d	LRISK	–	99.3	98.7	2000
zero_trades_252d	LRISK	–	97.6	93.6	2000
prc_highprc_252d	MOM	Yes	97.0	92.7	2000
resff3_12_1	MOM	Yes	90.6	80.5	2000
resff3_6_1	MOM	Yes	90.6	80.5	2000
ret_12_1	MOM	–	95.0	87.2	2000
ret_3_1	MOM	–	98.6	96.4	2021
ret_6_1	MOM	–	97.5	93.3	2000
ret_9_1	MOM	–	96.3	90.0	2000
seas_1_1na	MOM	–	95.3	87.8	2000
dsale_dinv	GROW	Yes	70.6	61.7	2022
dsale_drec	GROW	Yes	91.8	81.8	2000
dsale_dsga	GROW	Yes	77.0	65.6	2000

Table IA.1: Characteristic dictionary and availability (continued)

Characteristic	Theme	B39	Finite (%)	Worst (%)	Year
niq_at_chg1	GROW	–	94.9	87.7	2000
niq_be_chg1	GROW	–	90.2	80.5	2021
niq_su	GROW	–	92.2	81.2	2000
ocf_at_chg1	GROW	–	97.8	93.5	2022
ret_12_7	GROW	–	95.0	87.1	2000
sale_emp_gr1	GROW	–	90.6	77.0	2000
saleq_su	GROW	–	91.1	80.6	2000
seas_1_1an	GROW	–	95.3	87.8	2000
tax_gr1a	GROW	–	98.0	93.4	2022
dolvol_var_126d	PROF	Yes	98.7	96.9	2021
ebit_bev	PROF	Yes	95.4	89.4	2021
ebit_sale	PROF	Yes	97.9	92.7	2021
f_score	PROF	–	84.5	76.2	2000
ni_be	PROF	–	95.9	90.5	2021
niq_be	PROF	–	94.6	87.6	2021
o_score	PROF	–	88.5	84.2	2022
ocf_at	PROF	–	99.1	96.6	2021
ope_be	PROF	–	83.7	78.6	2000
ope_bel1	PROF	–	81.9	73.4	2000
turnover_var_126d	PROF	–	98.7	96.9	2021
at_turnover	QUAL	Yes	99.1	95.7	2021
cop_at	QUAL	Yes	96.9	90.3	2003
cop_atl1	QUAL	Yes	96.5	90.1	2003
dgp_dsale	QUAL	–	90.2	79.6	2000
gp_at	QUAL	–	98.9	96.2	2021
gp_atl1	QUAL	–	98.3	94.3	2021
mispricing_perf	QUAL	–	99.5	97.4	2021
ni_inc8q	QUAL	–	88.8	76.2	2000
niq_at	QUAL	–	98.3	94.2	2021
op_at	QUAL	–	99.2	96.6	2021
op_atl1	QUAL	–	98.5	94.6	2021
opex_at	QUAL	–	99.0	96.2	2021
qmj	QUAL	–	83.2	64.3	2000
qmj_growth	QUAL	–	83.2	64.3	2000
qmj_prof	QUAL	–	99.1	96.1	2021
qmj_safety	QUAL	–	99.4	97.2	2021
sale_bev	QUAL	–	95.5	89.4	2021
corr_1260d	SEAS	Yes	86.1	74.2	2000
coskew_21d	SEAS	Yes	99.0	98.3	2021
dbnetis_at	SEAS	Yes	99.2	96.4	2021
kz_index	SEAS	–	89.5	82.4	2021
lti_gr1a	SEAS	–	90.4	87.4	2022

Table IA.1: Characteristic dictionary and availability (continued)

Characteristic	Theme	B39	Finite (%)	Worst (%)	Year
pi_nix	SEAS	–	79.1	62.3	2021
seas_11_15an	SEAS	–	49.9	31.6	2000
seas_11_15na	SEAS	–	49.8	31.6	2000
seas_16_20an	SEAS	–	39.5	24.3	2000
seas_2_5an	SEAS	–	78.2	61.1	2000
seas_6_10an	SEAS	–	62.0	41.5	2000
sti_gr1a	SEAS	–	96.1	92.7	2022
iskew_capm_21d	REV	Yes	99.0	98.3	2021
iskew_ff3_21d	REV	Yes	99.0	98.3	2021
iskew_hxz4_21d	REV	Yes	99.0	98.3	2021
ret_1_0	REV	–	99.4	98.5	2021
rmax5_rvol_21d	REV	–	97.0	92.6	2000
rskew_21d	REV	–	99.0	98.3	2021
ami_126d	SIZE	Yes	98.5	96.4	2021
dolvol_126d	SIZE	Yes	98.8	96.9	2021
market_equity	SIZE	Yes	100.0	100.0	2000
prc	SIZE	–	100.0	100.0	2000
rd_me	SIZE	–	50.6	46.9	2007
at_me	VAL	Yes	99.4	97.5	2022
be_me	VAL	Yes	96.1	91.3	2021
bev_mev	VAL	Yes	95.5	90.0	2021
chcsho_12m	VAL	–	94.8	86.6	2000
debt_me	VAL	–	99.4	97.5	2022
div12m_me	VAL	–	95.3	88.0	2000
ebitda_mev	VAL	–	98.8	95.4	2022
eq_dur	VAL	–	90.8	81.1	2021
eqnetis_at	VAL	–	96.8	90.2	2003
eqnpo_12m	VAL	–	95.1	87.1	2000
eqnpo_me	VAL	–	96.3	90.1	2003
eqpo_me	VAL	–	90.8	80.3	2000
fcf_me	VAL	–	96.7	88.8	2002
ival_me	VAL	–	77.5	60.9	2021
netis_at	VAL	–	96.8	90.2	2003
ni_me	VAL	–	99.2	96.6	2021
ocf_me	VAL	–	99.1	96.6	2021
sale_me	VAL	–	99.2	96.6	2021

Table IA.2: Results for all linear specifications

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. B39 is the stratified 39-characteristic baseline; H36 the historical proxy; F153 the full library. B/Q mark theme-balanced/Gram penalties; D is selective duplication, O family omission and P random-path seed/size. p counts canonical slopes, so duplication rows retain $p = 39$. R^2 uses the zero-forecast denominator, in percent. Mean is the percentage of monthly fits selecting the historical mean. Δ loss is model minus B39, in percentage points of zero-forecast loss; negative favors the model. Omission rows here also use B39; the full-library comparison is separate. The 95 percent simultaneous bands cover 92 nontrivial B39 contrasts, using 2,000 paired circular 12-month bootstrap draws conditional on fitted decisions. Theme codes: ACCR = Accruals, DEBT = Debt Issuance, INV = Investment, LEV = Low Leverage, LRISK = Low Risk, MOM = Momentum, GROW = Profit Growth, PROF = Profitability, QUAL = Quality, SEAS = Seasonality, REV = Short-Term Reversal, SIZE = Size, VAL = Value.

ID	p	R^2 (%)	Mean (%)	Δ loss	Simultaneous band
B39	39	0.214	48.0	0.000	[0.000,0.000]
H36	36	0.203	43.0	0.011	[-0.074,0.095]
F153	153	0.206	36.3	0.008	[-0.091,0.106]
F153-B	153	0.207	36.9	0.007	[-0.094,0.108]
F153-Q	153	0.206	21.8	0.008	[-0.085,0.102]
D-ACCRx2	39	0.214	46.9	0.000	[-0.001,0.001]
D-ACCRx5	39	0.210	43.0	0.003	[-0.006,0.013]
D-ACCRx10	39	0.209	41.3	0.005	[-0.008,0.019]
O-ACCR	147	0.213	35.2	0.001	[-0.093,0.095]
D-DEBTx2	39	0.219	48.6	-0.005	[-0.022,0.011]
D-DEBTx5	39	0.223	47.5	-0.009	[-0.028,0.010]
D-DEBTx10	39	0.224	44.1	-0.010	[-0.035,0.014]
O-DEBT	146	0.206	38.0	0.007	[-0.082,0.097]
D-INVx2	39	0.220	45.8	-0.006	[-0.023,0.011]
D-INVx5	39	0.224	46.4	-0.010	[-0.035,0.015]
D-INVx10	39	0.222	46.9	-0.008	[-0.033,0.017]
O-INV	131	0.210	34.6	0.004	[-0.089,0.097]
D-LEVx2	39	0.217	49.2	-0.003	[-0.011,0.006]
D-LEVx5	39	0.209	50.8	0.005	[-0.018,0.028]
D-LEVx10	39	0.207	52.0	0.007	[-0.019,0.033]
O-LEV	142	0.203	33.0	0.011	[-0.089,0.110]
D-LRISKx2	39	0.208	52.5	0.006	[-0.012,0.024]
D-LRISKx5	39	0.210	53.6	0.004	[-0.015,0.023]
D-LRISKx10	39	0.210	55.3	0.004	[-0.021,0.029]
O-LRISK	135	0.209	29.1	0.005	[-0.072,0.082]
D-MOMx2	39	0.206	49.2	0.008	[-0.015,0.032]
D-MOMx5	39	0.205	50.3	0.009	[-0.017,0.035]

Table IA.2: Results for all linear specifications (continued)

ID	p	R^2 (%)	Mean (%)	Δ loss	Simultaneous band
D-MOMx10	39	0.202	50.3	0.012	[-0.018,0.041]
O-MOM	145	0.192	33.0	0.022	[-0.072,0.116]
D-GROWx2	39	0.216	48.6	-0.002	[-0.007,0.004]
D-GROWx5	39	0.216	49.7	-0.002	[-0.010,0.005]
D-GROWx10	39	0.217	54.2	-0.003	[-0.012,0.006]
O-GROW	141	0.191	33.5	0.023	[-0.082,0.128]
D-PROFx2	39	0.214	47.5	-0.001	[-0.005,0.004]
D-PROFx5	39	0.214	40.2	0.000	[-0.010,0.009]
D-PROFx10	39	0.215	33.5	-0.001	[-0.016,0.014]
O-PROF	142	0.190	37.4	0.024	[-0.110,0.159]
D-QUALx2	39	0.215	40.2	-0.001	[-0.008,0.005]
D-QUALx5	39	0.217	20.7	-0.003	[-0.018,0.012]
D-QUALx10	39	0.219	17.9	-0.005	[-0.024,0.013]
O-QUAL	136	0.180	44.7	0.034	[-0.109,0.176]
D-SEASx2	39	0.213	48.6	0.001	[-0.001,0.003]
D-SEASx5	39	0.223	52.5	-0.009	[-0.028,0.010]
D-SEASx10	39	0.225	52.0	-0.011	[-0.030,0.009]
O-SEAS	141	0.208	35.8	0.006	[-0.094,0.107]
D-REVx2	39	0.214	48.0	0.000	[-0.000,0.001]
D-REVx5	39	0.213	48.6	0.001	[-0.001,0.002]
D-REVx10	39	0.215	48.0	-0.001	[-0.008,0.005]
O-REV	147	0.204	36.9	0.010	[-0.091,0.111]
D-SIZEx2	39	0.221	47.5	-0.007	[-0.025,0.011]
D-SIZEx5	39	0.226	42.5	-0.012	[-0.034,0.009]
D-SIZEx10	39	0.227	40.2	-0.013	[-0.035,0.009]
O-SIZE	148	0.209	34.1	0.005	[-0.089,0.099]
D-VALx2	39	0.208	51.4	0.006	[-0.014,0.026]
D-VALx5	39	0.206	48.6	0.008	[-0.019,0.034]
D-VALx10	39	0.201	48.0	0.013	[-0.018,0.044]
O-VAL	135	0.205	29.6	0.009	[-0.097,0.114]
P01/13	13	0.252	24.0	-0.038	[-0.110,0.034]
P01/39	39	0.213	37.4	0.001	[-0.070,0.072]
P01/78	78	0.214	36.3	-0.001	[-0.088,0.087]
P02/13	13	0.216	30.7	-0.002	[-0.047,0.044]
P02/39	39	0.214	41.9	0.000	[-0.057,0.057]
P02/78	78	0.204	31.8	0.010	[-0.087,0.107]
P03/13	13	0.228	22.9	-0.014	[-0.070,0.042]
P03/39	39	0.222	43.6	-0.008	[-0.074,0.058]
P03/78	78	0.211	40.2	0.002	[-0.074,0.079]
P04/13	13	0.217	43.0	-0.003	[-0.040,0.033]
P04/39	39	0.205	38.5	0.009	[-0.063,0.081]
P04/78	78	0.196	36.9	0.018	[-0.076,0.113]

Table IA.2: Results for all linear specifications (continued)

ID	p	R^2 (%)	Mean (%)	Δ loss	Simultaneous band
P05/13	13	0.234	17.9	-0.020	[-0.097,0.057]
P05/39	39	0.209	28.5	0.005	[-0.057,0.067]
P05/78	78	0.213	29.6	0.001	[-0.070,0.072]
P06/13	13	0.231	33.0	-0.017	[-0.055,0.022]
P06/39	39	0.198	27.4	0.016	[-0.056,0.087]
P06/78	78	0.207	26.8	0.007	[-0.101,0.115]
P07/13	13	0.227	25.1	-0.013	[-0.067,0.040]
P07/39	39	0.211	32.4	0.003	[-0.067,0.074]
P07/78	78	0.200	35.2	0.014	[-0.114,0.142]
P08/13	13	0.241	25.7	-0.027	[-0.111,0.057]
P08/39	39	0.208	40.2	0.006	[-0.060,0.072]
P08/78	78	0.194	35.8	0.020	[-0.097,0.137]
P09/13	13	0.213	33.5	0.001	[-0.068,0.070]
P09/39	39	0.214	40.2	0.000	[-0.065,0.065]
P09/78	78	0.208	37.4	0.006	[-0.102,0.114]
P10/13	13	0.226	26.8	-0.012	[-0.087,0.064]
P10/39	39	0.213	35.8	0.001	[-0.078,0.080]
P10/78	78	0.204	29.6	0.010	[-0.113,0.133]
P11/13	13	0.210	48.6	0.004	[-0.063,0.071]
P11/39	39	0.209	33.0	0.005	[-0.079,0.089]
P11/78	78	0.207	34.6	0.007	[-0.091,0.105]
P12/13	13	0.226	33.5	-0.012	[-0.105,0.081]
P12/39	39	0.207	27.4	0.007	[-0.087,0.100]
P12/78	78	0.201	30.2	0.013	[-0.094,0.119]

Table IA.3: Selective-copy mechanism, 2 copies

Theme	Total RMS	Fixed λ	Retuning	Cross	Same λ (%)
Accruals	0.57	0.34	0.26	-0.04	94.4
Debt Issuance	0.57	0.30	0.27	0.02	97.8
Investment	1.30	0.48	0.90	-0.34	89.4
Low Leverage	1.30	0.57	0.82	-0.25	91.1
Low Risk	1.55	0.60	1.02	-0.17	90.5
Momentum	1.50	0.56	0.99	-0.08	90.5
Profit Growth	0.50	0.16	0.35	-0.01	97.8
Profitability	1.40	0.65	0.80	-0.18	93.3
Quality	2.08	0.96	1.28	-1.01	83.8
Seasonality	1.30	0.63	0.76	-0.45	91.6
Short-Term Reversal	0.19	0.18	0.02	0.00	98.9
Size	1.50	0.49	1.05	-0.16	88.8
Value	1.63	0.65	1.11	-0.44	87.2

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Copies counts the total representations of each characteristic in the selected baseline theme. Root mean squared (RMS) forecast differences are averaged over monthly cross sections, in basis points. Total compares the retuned copied fit with the original; Fixed λ holds the original selected penalty; Retuning compares copied fits with retuned and fixed penalties. Cross is the mean inner product of the latter two forecast changes, in squared basis points; the squared-distance identity includes twice this term. Same λ is the percentage of months retaining the baseline choice. RMS averages are not additive. Direct expanded-copy and induced canonical solutions agree to a maximum coefficient difference of 1.4×10^{-16} over 117 comparisons at three formation dates.

Table IA.4: Selective-copy mechanism, 5 copies

Theme	Total RMS	Fixed λ	Retuning	Cross	Same λ (%)
Accruals	1.27	0.90	0.48	-0.23	86.0
Debt Issuance	1.57	0.77	0.88	-0.05	92.2
Investment	2.96	1.16	2.25	-1.34	74.3
Low Leverage	2.75	1.49	1.55	-1.00	82.7
Low Risk	3.22	1.59	2.04	-1.14	83.8
Momentum	3.03	1.46	1.99	-1.63	81.0
Profit Growth	0.97	0.40	0.61	-0.04	93.9
Profitability	2.95	1.74	1.45	-0.90	80.4
Quality	4.68	2.36	3.26	-5.64	54.2
Seasonality	2.67	1.61	1.88	-3.97	82.1
Short-Term Reversal	0.59	0.47	0.12	-0.01	97.2
Size	2.60	1.23	1.56	-0.45	78.8
Value	3.03	1.61	2.06	-3.23	76.0

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Copies counts the total representations of each characteristic in the selected baseline theme. Root mean squared (RMS) forecast differences are averaged over monthly cross sections, in basis points. Total compares the retuned copied fit with the original; Fixed λ holds the original selected penalty; Retuning compares copied fits with retuned and fixed penalties. Cross is the mean inner product of the latter two forecast changes, in squared basis points; the squared-distance identity includes twice this term. Same λ is the percentage of months retaining the baseline choice. RMS averages are not additive. Direct expanded-copy and induced canonical solutions agree to a maximum coefficient difference of 1.4×10^{-16} over 117 comparisons at three formation dates.

Table IA.5: Selective-copy mechanism, 10 copies

Theme	Total RMS	Fixed λ	Retuning	Cross	Same λ (%)
Accruals	1.96	1.35	0.93	-1.01	81.0
Debt Issuance	2.14	1.15	1.15	-0.21	83.2
Investment	3.71	1.66	2.84	-2.36	68.7
Low Leverage	3.97	2.27	2.45	-3.18	73.7
Low Risk	3.46	2.41	1.87	-4.31	83.8
Momentum	3.91	2.20	2.60	-3.84	76.0
Profit Growth	1.49	0.61	0.98	-0.17	89.9
Profitability	4.35	2.67	2.17	-1.88	71.5
Quality	6.15	3.35	4.55	-12.35	44.7
Seasonality	3.29	2.40	2.48	-9.42	75.4
Short-Term Reversal	1.19	0.72	0.51	-0.05	93.9
Size	3.40	1.79	1.99	-0.86	72.1
Value	3.75	2.30	2.71	-7.72	69.8

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Copies counts the total representations of each characteristic in the selected baseline theme. Root mean squared (RMS) forecast differences are averaged over monthly cross sections, in basis points. Total compares the retuned copied fit with the original; Fixed λ holds the original selected penalty; Retuning compares copied fits with retuned and fixed penalties. Cross is the mean inner product of the latter two forecast changes, in squared basis points; the squared-distance identity includes twice this term. Same λ is the percentage of months retaining the baseline choice. RMS averages are not additive. Direct expanded-copy and induced canonical solutions agree to a maximum coefficient difference of 1.4×10^{-16} over 117 comparisons at three formation dates.

Table IA.6: All managed representations: Blocked ridge

Encoding	Share dist.	Gross SR	25 bp SR	Spread SR	Trades
Original	0.000	0.366	-0.083	-0.406	1.13
Balanced	0.231	0.348	-0.126	-0.467	1.19
Accruals x5	0.114	0.345	-0.122	-0.459	1.13
Debt Issuance x5	0.156	0.317	-0.118	-0.432	1.06
Investment x5	0.299	0.515	0.099	-0.202	1.03
Low Leverage x5	0.212	0.485	0.039	-0.286	1.07
Low Risk x5	0.281	0.501	-0.015	-0.376	1.31
Momentum x5	0.168	0.485	-0.007	-0.356	1.19
Profit Growth x5	0.125	0.284	-0.151	-0.451	1.14
Profitability x5	0.211	0.447	-0.041	-0.392	1.09
Quality x5	0.312	0.613	0.159	-0.187	1.04
Seasonality x5	0.176	0.426	-0.082	-0.408	1.33
Short-Term Reversal x5	0.098	0.313	-0.210	-0.577	1.36
Size x5	0.149	0.325	-0.108	-0.425	1.11
Value x5	0.299	0.254	-0.163	-0.462	1.09

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Share dist. is the mean half- ℓ_1 change in 13 training covariance shares relative to Original. Balanced uses theme-count scaling; x5 gives each named theme characteristic five total copies. SR is annualized Sharpe; Trades is mean full absolute trading per dollar of net asset value in the gross portfolio. Directions target 10 percent training volatility, then cap gross exposure at two. Cost scenarios use separate accounts: 25 basis points per traded dollar or a half-spread proxy clipped to 5–500 basis points. Missing returns use zero vendor excess returns; impact and borrow fees are omitted. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.7: All managed representations: SR-weighted

Encoding	Share dist.	Gross SR	25 bp SR	Spread SR	Trades
Original	0.000	0.490	0.001	-0.335	1.22
Balanced	0.198	0.397	-0.086	-0.421	1.24
Accruals x5	0.097	0.471	0.004	-0.321	1.16
Debt Issuance x5	0.181	0.468	0.015	-0.297	1.13
Investment x5	0.296	0.434	-0.054	-0.400	1.18
Low Leverage x5	0.175	0.442	-0.035	-0.361	1.19
Low Risk x5	0.278	0.394	-0.107	-0.447	1.28
Momentum x5	0.206	0.460	-0.017	-0.353	1.26
Profit Growth x5	0.126	0.419	-0.030	-0.346	1.17
Profitability x5	0.195	0.467	-0.056	-0.423	1.27
Quality x5	0.313	0.742	0.209	-0.186	1.08
Seasonality x5	0.154	0.407	-0.175	-0.533	1.43
Short-Term Reversal x5	0.082	0.485	-0.050	-0.420	1.35
Size x5	0.115	0.463	-0.011	-0.342	1.18
Value x5	0.267	0.370	-0.103	-0.437	1.24

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Share dist. is the mean half- ℓ_1 change in 13 training covariance shares relative to Original. Balanced uses theme-count scaling; x5 gives each named theme characteristic five total copies. SR is annualized Sharpe; Trades is mean full absolute trading per dollar of net asset value in the gross portfolio. Directions target 10 percent training volatility, then cap gross exposure at two. Cost scenarios use separate accounts: 25 basis points per traded dollar or a half-spread proxy clipped to 5–500 basis points. Missing returns use zero vendor excess returns; impact and borrow fees are omitted. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.8: All managed representations: LOO ridge

Encoding	Share dist.	Gross SR	25 bp SR	Spread SR	Trades
Original	0.000	1.003	0.152	-0.424	1.28
Balanced	0.271	1.105	0.190	-0.424	1.32
Accruals x5	0.116	1.069	0.205	-0.385	1.27
Debt Issuance x5	0.301	1.005	0.177	-0.391	1.19
Investment x5	0.347	1.079	0.221	-0.390	1.15
Low Leverage x5	0.262	1.046	0.264	-0.280	1.18
Low Risk x5	0.252	0.852	0.027	-0.535	1.36
Momentum x5	0.159	1.043	0.120	-0.520	1.33
Profit Growth x5	0.175	0.899	-0.024	-0.654	1.27
Profitability x5	0.326	1.136	0.349	-0.192	1.22
Quality x5	0.260	1.115	0.268	-0.334	1.19
Seasonality x5	0.212	1.023	-0.013	-0.668	1.44
Short-Term Reversal x5	0.105	1.134	0.165	-0.483	1.36
Size x5	0.120	1.008	0.188	-0.375	1.24
Value x5	0.223	1.069	0.237	-0.338	1.24

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Share dist. is the mean half- ℓ_1 change in 13 training covariance shares relative to Original. Balanced uses theme-count scaling; x5 gives each named theme characteristic five total copies. SR is annualized Sharpe; Trades is mean full absolute trading per dollar of net asset value in the gross portfolio. Directions target 10 percent training volatility, then cap gross exposure at two. Cost scenarios use separate accounts: 25 basis points per traded dollar or a half-spread proxy clipped to 5–500 basis points. Missing returns use zero vendor excess returns; impact and borrow fees are omitted. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.9: All managed representations: UPSA

Encoding	Share dist.	Gross SR	25 bp SR	Spread SR	Trades
Original	0.000	1.126	0.259	-0.336	1.23
Balanced	0.274	1.141	0.220	-0.404	1.28
Accruals x5	0.125	1.152	0.277	-0.324	1.22
Debt Issuance x5	0.291	1.148	0.323	-0.250	1.14
Investment x5	0.326	1.102	0.268	-0.324	1.12
Low Leverage x5	0.280	1.179	0.386	-0.169	1.15
Low Risk x5	0.274	1.058	0.248	-0.312	1.28
Momentum x5	0.170	1.121	0.218	-0.406	1.28
Profit Growth x5	0.185	1.004	0.100	-0.523	1.22
Profitability x5	0.297	1.211	0.377	-0.201	1.18
Quality x5	0.299	1.143	0.316	-0.272	1.17
Seasonality x5	0.211	1.071	0.078	-0.556	1.40
Short-Term Reversal x5	0.113	1.126	0.191	-0.439	1.33
Size x5	0.145	1.099	0.257	-0.324	1.20
Value x5	0.264	1.150	0.334	-0.234	1.19

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Share dist. is the mean half- ℓ_1 change in 13 training covariance shares relative to Original. Balanced uses theme-count scaling; x5 gives each named theme characteristic five total copies. SR is annualized Sharpe; Trades is mean full absolute trading per dollar of net asset value in the gross portfolio. Directions target 10 percent training volatility, then cap gross exposure at two. Cost scenarios use separate accounts: 25 basis points per traded dollar or a half-spread proxy clipped to 5–500 basis points. Missing returns use zero vendor excess returns; impact and borrow fees are omitted. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.10: IPCA convergence by refit year: 36-name proxy

Refit year	K1 R	K1 U	K3 R	K3 U	K5 R	K5 U
2011	7	7	18	19	69	137
2012	6	7	22	20	51	54
2013	6	6	34	33	82	66
2014	6	6	22	26	69	60
2015	6	6	28	28	38	32
2016	6	7	24	26	33	31
2017	6	7	22	25	26	28
2018	6	7	21	22	30	30
2019	7	7	28	25	253	259
2020	6	6	17	20	49	48
2021	6	6	22	26	53	57
2022	7	7	30	29	39	39
2023	7	7	27	27	47	52
2024	7	7	29	30	46	45
2025	7	7	29	30	43	39

Entries count alternating iterations for January fits of instrumented principal component analysis (IPCA). K is the factor count; R/U restricts/allows the residual characteristic intercept. Each fit uses 120 formation months ending two months before January, equal month weights, a constant characteristic and no observed factors. All met tolerance 10^{-6} before the 500-iteration limit; this does not establish global optimality. Final tolerances, orthogonality and complete-forecast transport errors accompany the supplementary estimates.

Table IA.11: IPCA convergence by refit year: 153-name library

Refit year	K1 R	K1 U	K3 R	K3 U	K5 R	K5 U
2011	7	7	37	25	47	49
2012	7	7	40	34	61	178
2013	6	6	43	55	104	92
2014	6	6	28	36	163	202
2015	6	6	33	28	70	75
2016	6	6	30	29	52	70
2017	6	6	34	35	37	42
2018	6	6	34	37	36	41
2019	6	7	34	28	86	92
2020	7	7	25	35	44	42
2021	6	6	36	56	58	76
2022	7	7	40	36	59	60
2023	7	7	37	36	57	63
2024	8	7	53	51	62	63
2025	7	7	55	57	88	142

Entries count alternating iterations for January fits of instrumented principal component analysis (IPCA). K is the factor count; R/U restricts/allows the residual characteristic intercept. Each fit uses 120 formation months ending two months before January, equal month weights, a constant characteristic and no observed factors. All met tolerance 10^{-6} before the 500-iteration limit; this does not establish global optimality. Final tolerances, orthogonality and complete-forecast transport errors accompany the supplementary estimates.

Table IA.12: Annual model selection: Gradient boosting with 39 characteristics

Year	Leaves	Iterations	Training rows	Validation loss
2011	0	0	255,375	0.02206
2012	0	0	247,504	0.01316
2013	0	0	243,690	0.01116
2014	0	0	241,446	0.00895
2015	0	0	241,848	0.00926
2016	0	0	244,098	0.01086
2017	0	0	245,770	0.01228
2018	7	100	247,004	0.01045
2019	0	0	248,237	0.01018
2020	15	50	250,215	0.01436
2021	0	0	254,347	0.02466
2022	0	0	263,199	0.03000
2023	0	0	273,652	0.02570
2024	15	100	279,926	0.02224
2025	0	0	281,677	0.01885

Each row is a January model selection on 120 training months; fitted parameters remain fixed that calendar year. Training rows counts stock-month observations in the outer fit. Validation loss is equal-month squared error in squared decimal-return units. Zero leaves/iterations selects the historical mean. The grid has 7/15 leaves, 50/100 iterations, learning rate 0.05, minimum leaf size 50 and L_2 penalty 10; early stopping is disabled. The 39- and 153-characteristic models together select the mean in 25 of 30 fits. This is the stated small-grid nonlinear comparison.

Table IA.13: Annual model selection: Gradient boosting with 153 characteristics

Year	Leaves	Iterations	Training rows	Validation loss
2011	0	0	255,375	0.02206
2012	0	0	247,504	0.01316
2013	0	0	243,690	0.01116
2014	0	0	241,446	0.00895
2015	0	0	241,848	0.00926
2016	0	0	244,098	0.01086
2017	7	50	245,770	0.01220
2018	7	50	247,004	0.01028
2019	0	0	248,237	0.01018
2020	0	0	250,215	0.01436
2021	0	0	254,347	0.02466
2022	0	0	263,199	0.03000
2023	0	0	273,652	0.02570
2024	0	0	279,926	0.02228
2025	0	0	281,677	0.01885

Each row is a January model selection on 120 training months; fitted parameters remain fixed that calendar year. Training rows counts stock-month observations in the outer fit. Validation loss is equal-month squared error in squared decimal-return units. Zero leaves/iterations selects the historical mean. The grid has 7/15 leaves, 50/100 iterations, learning rate 0.05, minimum leaf size 50 and L_2 penalty 10; early stopping is disabled. The 39- and 153-characteristic models together select the mean in 25 of 30 fits. This is the stated small-grid nonlinear comparison.

Table IA.14: Synthetic precision experiment, measurement noise 0.5

Design	m	Bayes MSE	Ridge MSE	Paired MC interval
Exact	1	0.4500	0.4864	[0.0000,0.0000]
Exact	2	0.4500	0.4826	[-0.0050,-0.0028]
Exact	4	0.4500	0.4777	[-0.0106,-0.0069]
Exact	8	0.4500	0.4717	[-0.0172,-0.0124]
Exact	16	0.4500	0.4666	[-0.0226,-0.0170]
Conditional	1	0.4500	0.4864	[0.0000,0.0000]
Conditional	2	0.4500	0.4958	[0.0072,0.0115]
Conditional	4	0.4500	0.4981	[0.0096,0.0138]
Conditional	8	0.4500	0.4910	[0.0018,0.0074]
Conditional	16	0.4500	0.4793	[-0.0109,-0.0034]
Independent	1	0.4500	0.4864	[0.0000,0.0000]
Independent	2	0.3611	0.3911	[-0.0978,-0.0929]
Independent	4	0.3088	0.3332	[-0.1564,-0.1501]
Independent	8	0.2803	0.2999	[-0.1899,-0.1832]
Independent	16	0.2654	0.2806	[-0.2094,-0.2022]

Synthetic Gaussian experiment with latent signal h , outcome $y = h + 0.5\epsilon$ and mutually independent standard Gaussian shocks. Exact repeats $x_1 = h + \tau e_1$; Conditional retains x_1 and adds $x_1 + \tau e_j$ for $j \geq 2$; Independent observes $h + \tau e_j$. The title gives noise standard deviation τ ; m includes the first measurement. Mean squared errors (MSE) average 100 replications with 120 training, 120 validation and 4,000 test observations and eight nuisance predictors. The paired Monte Carlo (MC) 95 percent interval is mean change from $m = 1$ plus/minus the 97.5th percentile of a Student t distribution with 99 degrees of freedom times its across-replication standard error, not an interval for the MSE level. Only independent measurements improve the Bayes information benchmark.

Table IA.15: Synthetic precision experiment, measurement noise 1

Design	m	Bayes MSE	Ridge MSE	Paired MC interval
Exact	1	0.7500	0.7999	[0.0000,0.0000]
Exact	2	0.7500	0.7894	[-0.0119,-0.0091]
Exact	4	0.7500	0.7805	[-0.0218,-0.0170]
Exact	8	0.7500	0.7736	[-0.0292,-0.0233]
Exact	16	0.7500	0.7695	[-0.0337,-0.0271]
Conditional	1	0.7500	0.7999	[0.0000,0.0000]
Conditional	2	0.7500	0.8077	[0.0055,0.0102]
Conditional	4	0.7500	0.8125	[0.0096,0.0155]
Conditional	8	0.7500	0.8054	[0.0010,0.0102]
Conditional	16	0.7500	0.7907	[-0.0142,-0.0043]
Independent	1	0.7500	0.7999	[0.0000,0.0000]
Independent	2	0.5833	0.6253	[-0.1782,-0.1711]
Independent	4	0.4500	0.4856	[-0.3190,-0.3096]
Independent	8	0.3611	0.3894	[-0.4152,-0.4057]
Independent	16	0.3088	0.3307	[-0.4745,-0.4638]

Synthetic Gaussian experiment with latent signal h , outcome $y = h + 0.5\epsilon$ and mutually independent standard Gaussian shocks. Exact repeats $x_1 = h + \tau e_1$; Conditional retains x_1 and adds $x_1 + \tau e_j$ for $j \geq 2$; Independent observes $h + \tau e_j$. The title gives noise standard deviation τ ; m includes the first measurement. Mean squared errors (MSE) average 100 replications with 120 training, 120 validation and 4,000 test observations and eight nuisance predictors. The paired Monte Carlo (MC) 95 percent interval is mean change from $m = 1$ plus/minus the 97.5th percentile of a Student t distribution with 99 degrees of freedom times its across-replication standard error, not an interval for the MSE level. Only independent measurements improve the Bayes information benchmark.

Table IA.16: Synthetic precision experiment, measurement noise 2

Design	m	Bayes MSE	Ridge MSE	Paired MC interval
Exact	1	1.0500	1.0853	[0.0000,0.0000]
Exact	2	1.0500	1.0767	[-0.0096,-0.0077]
Exact	4	1.0500	1.0721	[-0.0145,-0.0119]
Exact	8	1.0500	1.0688	[-0.0181,-0.0148]
Exact	16	1.0500	1.0668	[-0.0203,-0.0166]
Conditional	1	1.0500	1.0853	[0.0000,0.0000]
Conditional	2	1.0500	1.0916	[0.0039,0.0087]
Conditional	4	1.0500	1.0975	[0.0090,0.0153]
Conditional	8	1.0500	1.0917	[0.0036,0.0091]
Conditional	16	1.0500	1.0834	[-0.0047,0.0009]
Independent	1	1.0500	1.0853	[0.0000,0.0000]
Independent	2	0.9167	0.9532	[-0.1360,-0.1282]
Independent	4	0.7500	0.7890	[-0.3020,-0.2906]
Independent	8	0.5833	0.6254	[-0.4658,-0.4541]
Independent	16	0.4500	0.4910	[-0.5999,-0.5886]

Synthetic Gaussian experiment with latent signal h , outcome $y = h + 0.5\epsilon$ and mutually independent standard Gaussian shocks. Exact repeats $x_1 = h + \tau e_1$; Conditional retains x_1 and adds $x_1 + \tau e_j$ for $j \geq 2$; Independent observes $h + \tau e_j$. The title gives noise standard deviation τ ; m includes the first measurement. Mean squared errors (MSE) average 100 replications with 120 training, 120 validation and 4,000 test observations and eight nuisance predictors. The paired Monte Carlo (MC) 95 percent interval is mean change from $m = 1$ plus/minus the 97.5th percentile of a Student t distribution with 99 degrees of freedom times its across-replication standard error, not an interval for the MSE level. Only independent measurements improve the Bayes information benchmark.

Table IA.17: Fixed-asset alphas: Blocked ridge (1/2)

Asset	FF6	O	B	Asset	FF6	O	B
BM11	-17.2	-16.3	-16.5	BM51	14.6	14.9	15.0
BM12	13.2	14.8	14.9	BM52	2.6	2.7	2.7
BM13	4.9	5.3	5.0	BM53	-0.7	-1.1	-0.8
BM14	6.4	6.2	6.2	BM54	-6.7	-6.7	-6.5
BM15	46.6	47.4	47.3	BM55	11.9	12.1	11.9
BM21	9.9	10.0	10.1	OP11	-5.1	-3.7	-3.9
BM22	19.8	19.8	20.0	OP12	31.8	32.2	32.2
BM23	-4.4	-5.0	-5.1	OP13	34.2	34.4	34.3
BM24	6.9	6.1	6.1	OP14	60.3	60.8	60.9
BM25	-5.2	-5.0	-4.9	OP15	-32.7	-33.3	-33.3
BM31	-10.6	-11.1	-11.1	OP21	9.0	9.6	9.5
BM32	0.7	-0.3	0.0	OP22	-1.2	-1.3	-1.2
BM33	-13.4	-14.2	-14.1	OP23	15.3	14.7	14.8
BM34	-8.6	-8.8	-8.9	OP24	-0.5	-0.5	-0.3
BM35	-7.5	-7.4	-7.4	OP25	-21.6	-22.5	-22.6
BM41	-1.3	-2.3	-2.0	OP31	-4.7	-4.6	-4.4
BM42	-10.8	-11.6	-11.4	OP32	-10.5	-11.2	-11.3
BM43	-8.3	-8.4	-8.3	OP33	-5.4	-5.6	-5.7
BM44	-1.9	-2.2	-2.0	OP34	-5.5	-6.0	-5.9
BM45	-2.7	-2.7	-2.8				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.18: Fixed-asset alphas: Blocked ridge (2/2)

Asset	FF6	O	B	Asset	FF6	O	B
OP35	-22.4	-23.5	-23.2	IV25	0.4	1.5	1.6
OP41	-2.0	-1.9	-1.8	IV31	12.5	12.7	12.9
OP42	-9.6	-10.0	-10.0	IV32	-13.3	-13.6	-13.5
OP43	-2.8	-3.2	-3.1	IV33	-8.2	-9.0	-9.0
OP44	-5.8	-6.3	-6.2	IV34	-9.1	-9.0	-8.9
OP45	-5.1	-6.2	-5.8	IV35	-10.4	-11.2	-11.2
OP51	-13.2	-13.5	-13.8	IV41	-9.6	-10.1	-10.1
OP52	-4.6	-5.2	-5.1	IV42	-3.2	-4.1	-3.8
OP53	0.3	0.2	0.2	IV43	-7.2	-7.9	-7.8
OP54	10.8	11.8	11.8	IV44	-3.1	-3.6	-3.3
OP55	4.4	4.3	4.2	IV45	-2.1	-2.1	-2.0
IV11	20.7	20.9	20.9	IV51	-1.2	-0.7	-1.1
IV12	18.4	18.3	18.3	IV52	0.0	0.1	0.3
IV13	36.8	36.5	36.2	IV53	0.0	-0.1	-0.1
IV14	12.2	12.6	12.6	IV54	1.7	1.1	1.2
IV15	-3.9	-1.5	-1.7	IV55	13.9	14.5	14.7
IV21	-2.1	-2.5	-2.5				
IV22	-5.5	-6.3	-6.2				
IV23	9.6	9.1	9.0				
IV24	14.2	13.1	13.0				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.19: Fixed-asset alphas: SR-weighted (1/2)

Asset	FF6	O	B	Asset	FF6	O	B
BM11	-17.2	-15.0	-15.9	BM51	14.6	15.7	15.7
BM12	13.2	16.6	15.3	BM52	2.6	3.5	2.9
BM13	4.9	5.2	5.5	BM53	-0.7	-2.4	-1.8
BM14	6.4	7.3	7.1	BM54	-6.7	-6.1	-6.4
BM15	46.6	43.8	45.7	BM55	11.9	12.6	11.8
BM21	9.9	11.4	10.8	OP11	-5.1	-3.3	-3.5
BM22	19.8	20.1	20.4	OP12	31.8	33.9	33.0
BM23	-4.4	-8.2	-6.6	OP13	34.2	35.7	35.1
BM24	6.9	5.9	6.0	OP14	60.3	50.2	55.1
BM25	-5.2	-3.4	-3.9	OP15	-32.7	-33.6	-33.1
BM31	-10.6	-11.9	-11.8	OP21	9.0	9.9	9.4
BM32	0.7	-1.2	-0.6	OP22	-1.2	-0.6	-0.6
BM33	-13.4	-15.2	-14.8	OP23	15.3	14.3	14.8
BM34	-8.6	-9.3	-9.4	OP24	-0.5	0.5	0.3
BM35	-7.5	-5.5	-6.2	OP25	-21.6	-23.4	-23.2
BM41	-1.3	-3.1	-2.7	OP31	-4.7	-4.6	-4.7
BM42	-10.8	-13.3	-12.2	OP32	-10.5	-11.6	-11.6
BM43	-8.3	-8.7	-8.6	OP33	-5.4	-5.5	-5.6
BM44	-1.9	-1.4	-1.7	OP34	-5.5	-6.1	-6.0
BM45	-2.7	-0.5	-1.7				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.20: Fixed-asset alphas: SR-weighted (2/2)

Asset	FF6	O	B	Asset	FF6	O	B
OP35	-22.4	-25.5	-24.6	IV25	0.4	3.0	1.8
OP41	-2.0	-0.5	-1.7	IV31	12.5	12.1	11.9
OP42	-9.6	-9.2	-9.6	IV32	-13.3	-13.7	-13.5
OP43	-2.8	-3.3	-2.9	IV33	-8.2	-9.7	-9.5
OP44	-5.8	-7.7	-6.8	IV34	-9.1	-7.9	-8.4
OP45	-5.1	-8.2	-7.1	IV35	-10.4	-12.4	-12.1
OP51	-13.2	-14.6	-14.3	IV41	-9.6	-10.3	-10.1
OP52	-4.6	-6.1	-5.4	IV42	-3.2	-4.4	-3.8
OP53	0.3	0.8	0.6	IV43	-7.2	-8.5	-7.9
OP54	10.8	13.1	12.6	IV44	-3.1	-3.6	-3.4
OP55	4.4	4.2	4.3	IV45	-2.1	-2.2	-2.4
IV11	20.7	17.4	19.4	IV51	-1.2	0.8	0.0
IV12	18.4	18.2	18.7	IV52	0.0	1.1	0.8
IV13	36.8	36.1	36.0	IV53	0.0	-1.1	-0.6
IV14	12.2	14.4	13.6	IV54	1.7	0.2	0.8
IV15	-3.9	1.0	-0.7	IV55	13.9	16.4	16.1
IV21	-2.1	-3.6	-3.5				
IV22	-5.5	-6.0	-5.4				
IV23	9.6	8.5	9.0				
IV24	14.2	12.5	13.0				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.21: Fixed-asset alphas: LOO ridge (1/2)

Asset	FF6	O	B	Asset	FF6	O	B
BM11	-17.2	-17.8	-16.7	BM51	14.6	15.6	16.5
BM12	13.2	11.7	12.2	BM52	2.6	-2.0	-0.9
BM13	4.9	0.7	-0.3	BM53	-0.7	-2.0	-4.6
BM14	6.4	2.2	1.7	BM54	-6.7	-5.2	-7.1
BM15	46.6	49.8	49.9	BM55	11.9	5.2	4.8
BM21	9.9	9.0	7.2	OP11	-5.1	-2.8	-2.0
BM22	19.8	20.0	20.2	OP12	31.8	30.3	28.9
BM23	-4.4	-8.7	-8.3	OP13	34.2	28.0	26.8
BM24	6.9	3.9	2.6	OP14	60.3	59.9	64.0
BM25	-5.2	-6.1	-4.3	OP15	-32.7	-32.3	-33.3
BM31	-10.6	-15.0	-14.9	OP21	9.0	5.7	5.1
BM32	0.7	-0.5	0.9	OP22	-1.2	-5.3	-5.6
BM33	-13.4	-13.6	-13.3	OP23	15.3	14.9	14.9
BM34	-8.6	-9.8	-10.2	OP24	-0.5	5.3	6.2
BM35	-7.5	-13.5	-13.8	OP25	-21.6	-20.4	-20.8
BM41	-1.3	-1.5	-0.4	OP31	-4.7	-3.9	-2.9
BM42	-10.8	-8.7	-7.1	OP32	-10.5	-13.3	-13.4
BM43	-8.3	-5.7	-5.2	OP33	-5.4	-7.8	-8.3
BM44	-1.9	-4.0	-3.4	OP34	-5.5	-8.9	-8.2
BM45	-2.7	-2.1	-3.2				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.22: Fixed-asset alphas: LOO ridge (2/2)

Asset	FF6	O	B	Asset	FF6	O	B
OP35	-22.4	-25.8	-25.9	IV25	0.4	4.9	4.5
OP41	-2.0	-5.2	-3.8	IV31	12.5	8.5	8.8
OP42	-9.6	-8.6	-8.5	IV32	-13.3	-15.6	-14.8
OP43	-2.8	-1.2	-0.8	IV33	-8.2	-12.0	-12.4
OP44	-5.8	-2.9	-0.8	IV34	-9.1	-6.2	-4.4
OP45	-5.1	-5.1	-4.7	IV35	-10.4	-13.4	-13.7
OP51	-13.2	-12.6	-15.4	IV41	-9.6	-8.5	-6.2
OP52	-4.6	-6.2	-5.5	IV42	-3.2	-5.7	-5.3
OP53	0.3	2.5	3.1	IV43	-7.2	-8.1	-7.9
OP54	10.8	8.6	8.2	IV44	-3.1	-1.0	-0.3
OP55	4.4	3.7	4.2	IV45	-2.1	2.7	3.9
IV11	20.7	25.2	28.1	IV51	-1.2	-1.9	-2.9
IV12	18.4	14.6	13.4	IV52	0.0	0.9	0.4
IV13	36.8	30.9	30.7	IV53	0.0	-2.0	-2.3
IV14	12.2	8.4	8.2	IV54	1.7	-1.6	-1.8
IV15	-3.9	-1.2	-1.6	IV55	13.9	13.7	14.4
IV21	-2.1	-4.5	-2.5				
IV22	-5.5	-9.7	-9.8				
IV23	9.6	4.9	3.1				
IV24	14.2	6.9	6.0				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.23: Fixed-asset alphas: UPSA (1/2)

Asset	FF6	O	B	Asset	FF6	O	B
BM11	-17.2	-18.2	-16.2	BM51	14.6	16.3	16.7
BM12	13.2	13.1	13.7	BM52	2.6	-2.4	-1.7
BM13	4.9	-0.3	-0.1	BM53	-0.7	-3.7	-5.8
BM14	6.4	1.1	1.1	BM54	-6.7	-5.0	-6.2
BM15	46.6	50.5	51.6	BM55	11.9	2.1	1.2
BM21	9.9	10.9	10.3	OP11	-5.1	-2.1	-0.8
BM22	19.8	18.9	19.3	OP12	31.8	30.1	29.0
BM23	-4.4	-10.1	-9.5	OP13	34.2	26.8	25.7
BM24	6.9	2.3	0.8	OP14	60.3	62.3	66.6
BM25	-5.2	-4.6	-2.3	OP15	-32.7	-32.5	-32.0
BM31	-10.6	-14.9	-16.2	OP21	9.0	6.7	6.7
BM32	0.7	-1.6	-0.6	OP22	-1.2	-5.4	-5.6
BM33	-13.4	-15.1	-15.4	OP23	15.3	14.6	14.5
BM34	-8.6	-10.3	-10.7	OP24	-0.5	5.1	5.2
BM35	-7.5	-12.6	-12.5	OP25	-21.6	-22.2	-20.7
BM41	-1.3	-1.2	-0.3	OP31	-4.7	-2.4	-2.7
BM42	-10.8	-9.4	-8.4	OP32	-10.5	-15.0	-15.9
BM43	-8.3	-5.8	-6.0	OP33	-5.4	-8.2	-9.0
BM44	-1.9	-4.8	-4.2	OP34	-5.5	-9.2	-8.0
BM45	-2.7	-2.0	-1.8				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.24: Fixed-asset alphas: UPSA (2/2)

Asset	FF6	O	B	Asset	FF6	O	B
OP35	-22.4	-27.0	-27.0	IV25	0.4	5.4	5.2
OP41	-2.0	-5.2	-4.0	IV31	12.5	7.3	7.5
OP42	-9.6	-8.6	-8.6	IV32	-13.3	-16.6	-16.5
OP43	-2.8	-1.4	-0.6	IV33	-8.2	-13.2	-13.6
OP44	-5.8	-2.6	-1.6	IV34	-9.1	-5.6	-5.3
OP45	-5.1	-6.1	-6.0	IV35	-10.4	-13.0	-13.4
OP51	-13.2	-15.5	-16.2	IV41	-9.6	-8.5	-6.9
OP52	-4.6	-7.1	-6.5	IV42	-3.2	-6.4	-5.6
OP53	0.3	3.4	3.8	IV43	-7.2	-8.2	-7.8
OP54	10.8	8.8	9.5	IV44	-3.1	-0.5	-0.4
OP55	4.4	3.8	3.5	IV45	-2.1	2.2	3.2
IV11	20.7	26.5	29.9	IV51	-1.2	-1.9	-2.3
IV12	18.4	14.6	14.5	IV52	0.0	1.3	0.6
IV13	36.8	29.6	29.6	IV53	0.0	-1.9	-2.8
IV14	12.2	7.8	8.0	IV54	1.7	-3.0	-3.0
IV15	-3.9	-0.7	-1.3	IV55	13.9	14.7	16.1
IV21	-2.1	-4.1	-1.8				
IV22	-5.5	-9.7	-9.2				
IV23	9.6	3.5	1.3				
IV24	14.2	4.8	5.3				

Monthly regression intercepts in basis points, February 2011–December 2025 (179 months). FF6 uses market, size, value, profitability, investment and momentum factors. O/B adds the original/theme-balanced gross strategy return to FF6. BM, OP and IV are size–book-to-market, size–operating-profitability and size–investment portfolios; digits give size/characteristic quintiles, lowest = 1. Parts 1 and 2 together contain all 75 assets. Supplementary estimates include Newey–West errors (12 lags) and Holm probabilities across 75 intercepts per specification. These are conditional time-series alphas. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.25: Economic copy contrasts with family bands (1/3)

Method	Theme	Δ annual mean	Simultaneous band
SRW	Accruals	0.02	[-1.27,1.32]
SRW	Debt Issuance	0.10	[-2.12,2.33]
SRW	Investment	-0.40	[-2.68,1.87]
SRW	Low Leverage	-0.27	[-1.83,1.28]
SRW	Low Risk	-0.83	[-3.01,1.35]
SRW	Momentum	-0.14	[-2.00,1.71]
SRW	Profit Growth	-0.25	[-1.48,0.99]
SRW	Profitability	-0.42	[-2.69,1.85]
SRW	Quality	1.27	[-2.43,4.97]
SRW	Seasonality	-1.30	[-2.35,-0.25]
SRW	Short-Term Reversal	-0.39	[-2.02,1.24]
SRW	Size	-0.09	[-0.88,0.70]
SRW	Value	-0.82	[-2.68,1.04]
LOO	Accruals	0.22	[-0.64,1.07]
LOO	Debt Issuance	0.08	[-1.25,1.42]
LOO	Investment	0.20	[-1.12,1.53]
LOO	Low Leverage	0.52	[-0.64,1.68]
LOO	Low Risk	-0.55	[-2.12,1.02]

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each row compares five total copies of the named theme with Original within the same method. Annual mean differences are percentage points after 25 basis points per traded dollar. The 95 percent bands use 2,000 paired circular 12-month bootstrap draws and maximum standardized deviations across 13 family copies within each method. This family does not cover all methods, cost assumptions or research choices jointly. BR denotes blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage.

Table IA.26: Economic copy contrasts with family bands (2/3)

Method	Theme	Δ annual mean	Simultaneous band
LOO	Momentum	-0.17	[-1.13,0.79]
LOO	Profit Growth	-0.78	[-1.91,0.34]
LOO	Profitability	0.96	[-0.78,2.70]
LOO	Quality	0.44	[-0.69,1.57]
LOO	Seasonality	-0.74	[-1.76,0.28]
LOO	Short-Term Reversal	0.01	[-0.93,0.95]
LOO	Size	0.17	[-0.65,0.99]
LOO	Value	0.37	[-0.46,1.20]
UPSA	Accruals	0.06	[-0.33,0.46]
UPSA	Debt Issuance	0.23	[-0.42,0.89]
UPSA	Investment	-0.02	[-0.85,0.81]
UPSA	Low Leverage	0.58	[-0.17,1.34]
UPSA	Low Risk	0.08	[-0.88,1.04]
UPSA	Momentum	-0.17	[-0.61,0.26]
UPSA	Profit Growth	-0.70	[-1.52,0.13]
UPSA	Profitability	0.50	[-0.40,1.40]
UPSA	Quality	0.24	[-0.42,0.90]
UPSA	Seasonality	-0.77	[-1.47,-0.08]

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each row compares five total copies of the named theme with Original within the same method. Annual mean differences are percentage points after 25 basis points per traded dollar. The 95 percent bands use 2,000 paired circular 12-month bootstrap draws and maximum standardized deviations across 13 family copies within each method. This family does not cover all methods, cost assumptions or research choices jointly. BR denotes blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage.

Table IA.27: Economic copy contrasts with family bands (3/3)

Method	Theme	Δ annual mean	Simultaneous band
UPSA	Short-Term Reversal	-0.29	[-0.69,0.12]
UPSA	Size	0.00	[-0.41,0.40]
UPSA	Value	0.37	[-0.18,0.92]
BR	Accruals	-0.26	[-2.33,1.81]
BR	Debt Issuance	-0.24	[-2.17,1.70]
BR	Investment	1.37	[-1.33,4.07]
BR	Low Leverage	0.91	[-1.43,3.26]
BR	Low Risk	0.52	[-1.57,2.61]
BR	Momentum	0.58	[-1.52,2.68]
BR	Profit Growth	-0.56	[-2.35,1.23]
BR	Profitability	0.35	[-1.84,2.55]
BR	Quality	1.73	[-2.51,5.97]
BR	Seasonality	-0.02	[-1.80,1.77]
BR	Short-Term Reversal	-1.01	[-2.98,0.96]
BR	Size	-0.20	[-2.42,2.02]
BR	Value	-0.65	[-2.80,1.50]

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each row compares five total copies of the named theme with Original within the same method. Annual mean differences are percentage points after 25 basis points per traded dollar. The 95 percent bands use 2,000 paired circular 12-month bootstrap draws and maximum standardized deviations across 13 family copies within each method. This family does not cover all methods, cost assumptions or research choices jointly. BR denotes blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage.

Table IA.28: Different attribution games (1/2)

Method	Month	Game distance	Min. projection	Min. covariance
SRW O	201101	0.393	0.007	-0.098
SRW O	201806	0.239	0.006	0.000
SRW O	202511	0.445	0.007	-0.128
SRW B	201101	0.319	0.006	-0.111
SRW B	201806	0.162	0.002	0.000
SRW B	202511	0.421	0.010	-0.117
LOO O	201101	0.299	0.019	-0.096
LOO O	201806	0.584	0.028	-0.077
LOO O	202511	0.311	0.008	-0.059
LOO B	201101	0.305	0.012	-0.079
LOO B	201806	0.583	0.029	-0.078
LOO B	202511	0.311	0.011	-0.075

Month is formation date (YYYYMM); each covariance uses the preceding 120-month training window with the stated embargo. O/B denotes original/theme-balanced. Game distance is half the ℓ_1 difference between the 13 projection-Shapley and covariance shares; Min. columns are the smallest share under each allocation. Projection enumerates all 8,192 coalitions of scalar theme payoffs using a pseudoinverse cutoff of 10^{-10} . Covariance shares are Shapley allocations of the variance-of-sum game and may be negative. Coefficients are fixed in both games. BR is blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage.

Table IA.29: Different attribution games (2/2)

Method	Month	Game distance	Min. projection	Min. covariance
UPSA O	201101	0.332	0.018	-0.097
UPSA O	201806	0.460	0.017	-0.104
UPSA O	202511	0.230	0.007	-0.036
UPSA B	201101	0.383	0.014	-0.093
UPSA B	201806	0.561	0.014	-0.119
UPSA B	202511	0.229	0.011	-0.017
BR O	201101	0.312	0.019	-0.057
BR O	201806	0.177	0.007	-0.013
BR O	202511	0.239	0.008	-0.014
BR B	201101	0.167	0.014	-0.022
BR B	201806	0.127	0.010	-0.014
BR B	202511	0.187	0.011	-0.019

Month is formation date (YYYYMM); each covariance uses the preceding 120-month training window with the stated embargo. O/B denotes original/theme-balanced. Game distance is half the ℓ_1 difference between the 13 projection-Shapley and covariance shares; Min. columns are the smallest share under each allocation. Projection enumerates all 8,192 coalitions of scalar theme payoffs using a pseudoinverse cutoff of 10^{-10} . Covariance shares are Shapley allocations of the variance-of-sum game and may be negative. Coefficients are fixed in both games. BR is blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage.

Table IA.30: Theme shares: original and balanced

Panel A: Blocked ridge and SR-weighted ridge

Theme	BR O	BR B	SRW O	SRW B
Accruals	0.01	0.03	0.02	0.05
Debt Issuance	0.03	0.05	0.05	0.09
Investment	0.15	0.08	0.14	0.09
Low Leverage	0.05	0.06	0.04	0.04
Low Risk	0.13	0.10	0.08	0.07
Momentum	0.04	0.08	0.03	0.06
Profit Growth	-0.01	0.00	0.00	0.00
Profitability	0.06	0.08	0.06	0.08
Quality	0.26	0.20	0.35	0.28
Seasonality	0.06	0.06	0.07	0.07
Short-Term Reversal	0.01	0.03	0.01	0.02
Size	0.03	0.08	0.02	0.05
Value	0.19	0.16	0.12	0.11

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Entries are mean training covariance shares, as fractions, for the original (O) and theme-balanced (B) representations. Each share is the covariance of a fitted theme payoff with total payoff divided by total variance. Shares may be negative; unrounded means sum to one. BR is blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage. The two panels use the same 13 themes and include all four methods.

Table IA.30: Theme shares: original and balanced (continued)

Panel B: LOO ridge and UPSA

Theme	LOO O	LOO B	UPSA O	UPSA B
Accruals	0.01	0.04	0.01	0.04
Debt Issuance	0.08	0.16	0.07	0.14
Investment	0.16	0.07	0.16	0.08
Low Leverage	0.06	0.05	0.06	0.06
Low Risk	0.12	0.07	0.14	0.10
Momentum	0.07	0.10	0.08	0.09
Profit Growth	-0.06	-0.06	-0.08	-0.07
Profitability	0.13	0.14	0.12	0.13
Quality	0.27	0.21	0.32	0.25
Seasonality	0.10	0.10	0.08	0.08
Short-Term Reversal	0.01	0.02	0.01	0.03
Size	0.02	0.05	0.00	0.04
Value	0.02	0.03	0.03	0.05

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Entries are mean training covariance shares, as fractions, for the original (O) and theme-balanced (B) representations. Each share is the covariance of a fitted theme payoff with total payoff divided by total variance. Shares may be negative; unrounded means sum to one. BR is blocked ridge; SRW the training-Sharpe-weighted adaptation; LOO leave-one-out-selected ridge; UPSA Universal Portfolio Shrinkage. The two panels use the same 13 themes and include all four methods.

Table IA.31: Sensitivity to the rank tie convention

Rule	Share dist.	Weight dist.	Δ SR	Δ mean	Max. month
Blocked ridge O	0.00010	0.00020	0.00001	-0.00004	0.29
Blocked ridge B	0.00008	0.00021	0.00000	-0.00009	0.25
SR-weighted O	0.00021	0.00037	-0.00117	-0.00914	10.55
SR-weighted B	0.00018	0.00037	-0.00040	-0.00256	3.69
LOO ridge O	0.00211	0.00375	-0.00095	-0.00132	16.02
LOO ridge B	0.00043	0.00067	0.00005	0.00001	0.34
UPSA O	0.00060	0.00068	-0.00006	-0.00015	0.38
UPSA B	0.00063	0.00077	0.00003	0.00009	0.45

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Exact and baseline rank policies are compared after a full panel rebuild and separate refits. The baseline groups adjacent sorted values within 10^{-13} ; the alternative groups exact numeric ties. O/B denotes original/theme-balanced. Share dist. is mean half- ℓ_1 change in covariance shares; Weight dist. is mean ℓ_1 target-weight change divided by average gross exposure. Δ SR is exact minus baseline annualized gross Sharpe. Δ mean is the exact-minus-baseline annual mean change in percentage points at 25 basis points per traded dollar; Max. month is the largest absolute monthly return change in basis points under that scenario. These comparisons have no new confidence intervals; IPCA and boosting are excluded. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.32: Theme balancing under both rank conventions

Method	Share N	Share E	Stock N	Stock E	Mean N	Mean E
Blocked ridge	0.2308	0.2309	0.2773	0.2773	-0.3246	-0.3246
SR-weighted	0.1976	0.1976	0.2389	0.2389	-0.6734	-0.6668
LOO ridge	0.2710	0.2719	0.2269	0.2294	0.1362	0.1375
UPSA	0.2737	0.2737	0.1906	0.1906	-0.1863	-0.1860

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. Each entry compares the balanced and original representations within one rank policy. N is the near-tie baseline (10^{-13}); E uses exact numeric ties. Share is mean half- ℓ_1 covariance-share distance; Stock is mean ℓ_1 target-weight distance divided by average gross exposure. Mean is the balanced-minus-original annual mean excess-return change, in percentage points, at 25 basis points per traded dollar. Each policy separately refits the models. These sensitivity estimates have no new confidence intervals. SR-weighted denotes the training-Sharpe-weighted ridge adaptation; LOO ridge is leave-one-out-selected ridge; UPSA is Universal Portfolio Shrinkage.

Table IA.33: Two sections of the same unrestricted IPCA fit

Library	K	Factor I	Factor Q	Complete	Realized fit
36 proxy	1	-0.266	0.240	0.240	19.39
36 proxy	3	-0.018	0.267	0.302	22.42
36 proxy	5	0.135	0.298	0.299	23.57
153	1	-0.091	0.241	0.115	19.40
153	3	-0.246	0.224	0.193	22.48
153	5	-0.285	0.214	0.187	23.78

The sample contains 179 monthly formation decisions from January 2011 through November 2025, with returns realized from February 2011 through December 2025. IPCA denotes instrumented principal component analysis. Each model is refitted each January on 120 training months; K is the factor count. The 36-name proxy and full 153-name library use current Jensen–Kelly–Pedersen characteristic definitions. All entries are zero-forecast R^2 in percent. Factor I and Factor Q omit the residual intercept under, respectively, Euclidean and training-Gram decompositions. Complete includes the matched intercept and factor mean and is identical within each fit. Realized fit uses contemporaneous factors extracted from the return being fitted and is not a forecast. The libraries are separately estimated.

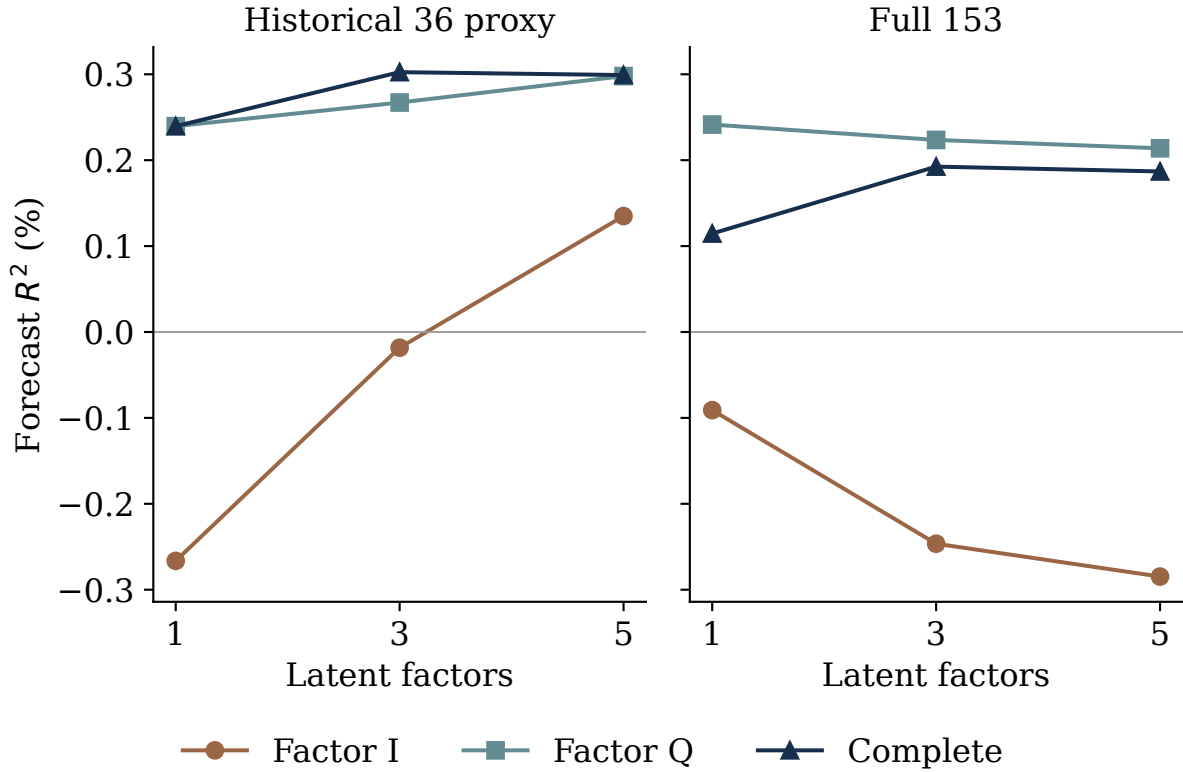


Figure IA.1. Section-dependent factor attribution and complete forecasts

The sample contains 179 formation decisions from January 2011 through November 2025, with returns through December 2025. Each instrumented principal component analysis (IPCA) model is refitted each January on 120 training months. The horizontal axis is the latent factor count; the vertical axis is forecast R^2 against zero, in percent. Factor I and Factor Q omit the residual intercept under Euclidean and training-Gram decompositions. Complete retains the matched intercept and factor mean and is identical within each fit. The historical 36-name proxy and full 153-name library are separately estimated; their complete forecasts need not coincide.